CLIM1001 wk1-3 key points

Introduction to Climate Change (University of New South Wales)

CLIM1001 Key points

Week1

Psychological Distance

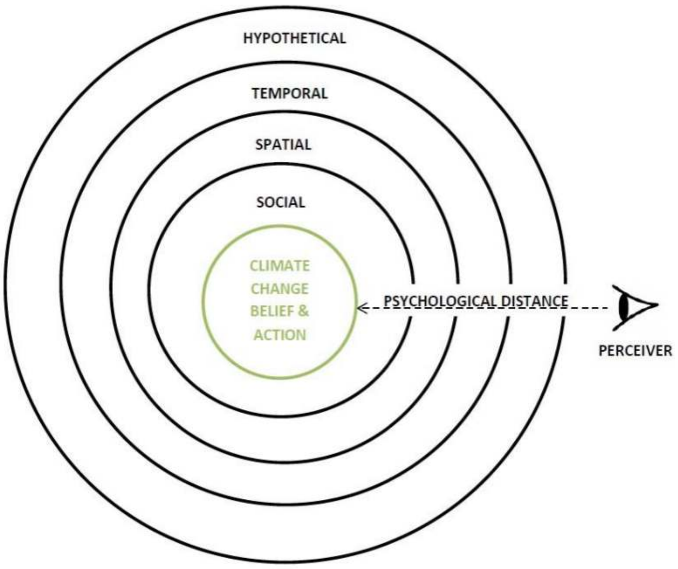
Climate change is a difficult problem for people to deal with and is one that many fail to understand. Despite the fact that climate change is a human induced problem that has the capacity to impact millions of lives, many people either deny climate change or prioritise it as an issue less important than war, terrorism and the economy. In this lesson, we will learn about why this is the case and the factors that make climate change such a difficult problem for people to engage with.

One way we can understand people's lack of concern over climate change is to use the concept of psychological distance. At its core, Psychological distance is a way of distancing oneself from a problem by refuting its hypothetical, temporal, spatial or social components.

心理距离

气候变化是人们难以应对的问题，也是许多人无法理解的问题。尽管气候变化是一个由人类引发的问题，有能力影响数百万人的生活，许多人要么否认气候变化，要么将其视为一个不如战争、恐怖主义和经济重要的问题。在这节课中，我们将了解为什么会出现这种情况，以及使气候变化成为人们难以应对的问题的因素。

我们可以通过心理距离的概念来理解人们对气候变化的不关注。在其核心，心理距离是一种远离问题的方式，通过驳斥其假设的、时间的、空间的或社会的成分。



Let's break the model down into its components so we gain a better understanding of how psychological distance can impact engagement with climate change:

- **Hypothetical distance**: Refers to the *hypothetical* element of a problem. For example, a person may ask themselves if climate change is an issue that's actually happening.
- **Temporal**: Refers to *when* a problem will occur. The person may ask themselves when the impacts of climate change will be felt.

- **Spatial**: Refers to *where* a problem will occur. The person may ask themselves where the impacts of climate change will be felt.
- **Social**: Refers to whether the problem will have a *personal* impact. The person may ask themselves whether climate change will personally affect them or those in their social group.

Putting these components together we can begin to understand the impact of psychological distance on climate change engagement. For example, even if an individual accepts that climate change is happening, they may still believe that its impacts will occur much later in the future, in a far-off place with limited impact on them personally. This example is precisely one of the main reasons that climate change is such a difficult problem to deal with. Climate change is often talked about in uncertain terms (e.g., is climate change happening?) and as something that will happen in the future (e.g., climate change will affect our future generations) with impacts far away (e.g., warming temperatures in Europe) that won't personally affect us (e.g., climate change impacts on sea levels will mostly impact small islands). As a result, people often underweight the importance of climate change and their personal contributions to the problem.

让我们把这个模型分解成各个部分，以便更好地理解心理距离如何影响对气候变化的参与:

假设距离:指问题中的假设元素。例如，一个人可能会问自己气候变化是否真的在发生。

暂时的:指何时会发生问题。人们可能会问自己什么时候会感受到气候变化的影响。

空间的:指问题将发生的地方。人们可能会问自己气候变化的影响会在哪里。

社会性:指问题是否会对个人产生影响。人们可能会问自己，气候变化是否会对他们个人或他们的社会群体产生影响。

把这些组成部分放在一起，我们就可以开始理解心理距离对气候变化参与的影响。例如，即使一个人接受气候变化正在发生，他们可能仍然相信它的影响将在遥远的未来发生，在一个对他们个人影响有限的地方。这个例子恰恰是气候变化是一个如此难以解决的问题的主要原因之一。气候变化经常被用不确定的术语来谈论(例如，气候变化正在发生吗?)，以及作为一种将在未来发生的事情(例如，气候变化将影响我们的后代)，其影响遥远(例如，欧洲的气温变暖)，而不会对我们个人产生影响(例如，气候变化对海平面的影响将主要影响小岛屿)。因此，人们往往低估了气候变化的重要性和他们个人对这个问题的贡献。

Risk Perception and Climate Change

Aside from Psychological Distance, there are a number of other processes that influence whether people engage with the problem of climate change. One significant contributor is the concept of risk perception. That is, how much of a risk people perceive climate change to be. Risk perception is important because people tend to be less willing to act when they don't perceive a problem to be a significant risk.

There are a number of components that influence people's perception of climate change as a risk. One model we can look at is the risk perception of climate change model (RPCC) by Sander van der Linden (2015). Within this model, there are a number of major components that contribute to risk perception such as cognitive factors, socio-cultural influences, experiential processing and socio-demographics. Breaking this down further:

- **Cognitive factors**: Our knowledge about the causes of climate change

- **Experiential processing**: Our personal experiences of climate change and our emotions towards the phenomenon
- **Socio-cultural influences**: What we are told to think or believe about climate change and how much we identify as an environmentally minded person

风险感知与气候变化

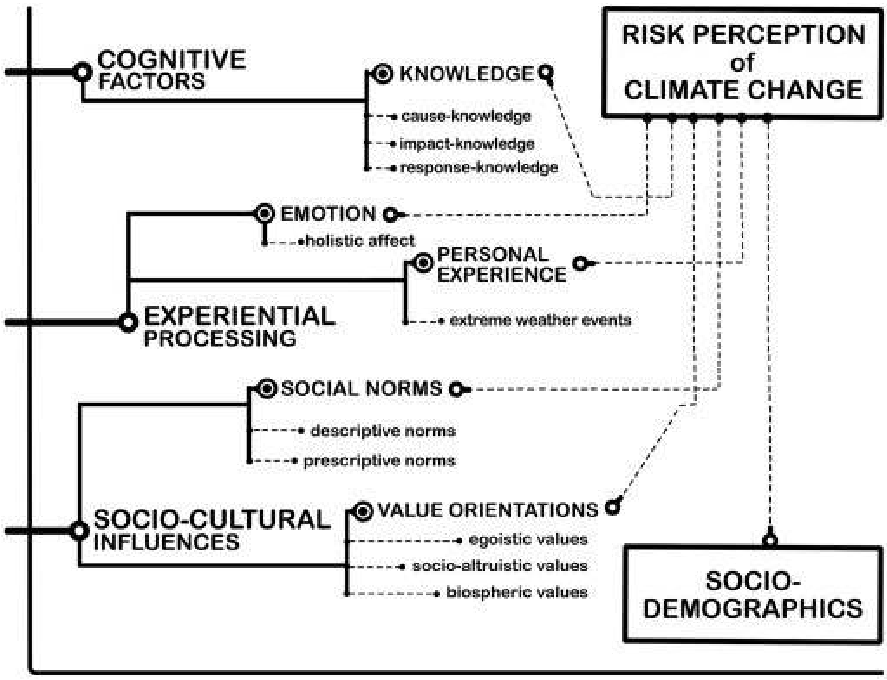
除了心理距离之外，还有许多其他过程会影响人们是否参与气候变化问题。一个重要的贡献者是**风险感知**的概念。也就是说，人们认为气候变化有多大的风险。风险认知很重要，因为当人们不认为一个问题是重大风险时，他们往往不太愿意采取行动。

有许多因素影响着人们对气候变化风险的看法。我们可以参考的一个模型是 Sander van der Linden(2015)的气候变化风险感知模型(RPCC)。在这个模型中，有许多主要因素有助于风险感知，如认知因素、社会文化影响、经验处理和社会人口统计。进一步细分如下:

认知因素:我们对气候变化原因的认识

经验处理:我们对气候变化的个人经历和我们对这一现象的情绪

社会文化影响:我们被告知应该如何看待或相信气候变化，以及我们在多大程度上认为自己是一个有环境意识的人



In a questionnaire based study, van der Linden (2015) surveyed 800 UK residents and asked them a number of questions about climate change. The data from the study suggested that experiential processing played the largest role in influencing people's risk perception of climate change. In fact, people's emotional responses towards climate change was the greatest predictor of whether people would perceive climate change as a risk. That is, our personal views of climate change as a good or bad phenomenon is what influences our perception of climate change risk the most.

Risk perception can be a complicated concept to deal with and we're not expecting you to memorise this figure. Instead, the main point that you should be getting from this section is that the way people perceive climate change risk is complicated and that people rely predominantly on experiential processing when they evaluate whether climate change is a risk/problem.

在一项基于问卷的研究中，van der Linden(2015)调查了 800 名英国居民，问了他们一些关于气候变化的问题。研究数据表明，体验加工在影响人们对气候变化的风险感知方面发挥了最大的作用。事实上，人们对气候变化的情绪反应是人们是否将气候变化视为一种风险的最大预测器。也就是说，**我们个人对气候变化好坏的看法是影响我们对气候变化风险感知的最大因素。**

风险感知是一个复杂的概念，我们不希望你记住这个数字。相反，你应该从这部分得到的主要观点是，人们感知气候变化风险的方式是复杂的，人们在评估气候变化是否是个风险/问题时主要依赖经验处理。

The problem with experiential processing

There is a large problem with using our personal experiences to evaluate the risk of climate change and that is that *we don't experience climate*. We experience much shorter term events - weather (we'll talk more about this distinction later). The fact that we don't experience climate can provide huge barriers for our ability to correctly understand climate change as a risk.
As pointed out by climate scientist James Hansen "*The greatest barrier to public recognition of human-made climate change is the natural variability of climate. How can a person discern long-term climate change, given the notorious variability of local weather and climate from day to day and year to year?*".

There are numerous ways of how relying too much on experience can lead to confusion or doubt about climate change. We can see an example of this in 80 year old regular beach-goer Kevin Court's statement about climate change and rising sea levels as quoted in The Australian newspaper:

> "I have swum at this beach every day for the past 50 years, and nothing much changes here," Mr Court said yesterday as he emerged from the surf at Wollongong's North Beach, just a short paddle from the Port Kembla gauging station.
>
> "All this talk about rising sea levels - most of us old-timers haven't seen any change and we've been coming down here for decades."

From Kevin's perspective, the sea level hasn't appeared to have risen in the last few decades that he has been going to the beach. Naturally, he is unconcerned about the effect of climate change on rising sea levels because he doesn't perceive any difference in the sea levels he's used to. However, this doesn't mean that sea levels haven't been rising.
If we think about other sources of data aside from our personal experience, we can see that the real story can be quite different. The graph provided below illustrates this point and shows that sea levels have been consistently rising since the late 1800's.

经验处理的问题

用我们的个人经验来评估气候变化的风险存在一个大问题，那就是我们没有经历气候变化。我们经历了更短期的事件——天气(我们稍后会详细讨论这一区别)。我们没有经历过气候变化，这一事实为我们正确理解气候变化是一种风险的能力提供了巨大的障碍。正如气候学家詹姆斯·

汉森(James Hansen)指出的那样:“公众认识到人为气候变化的最大障碍是气候的自然变化。”考虑到当地天气和气候日复一日、年复一年的变化无常，一个人如何能够辨别长期的气候变化?”

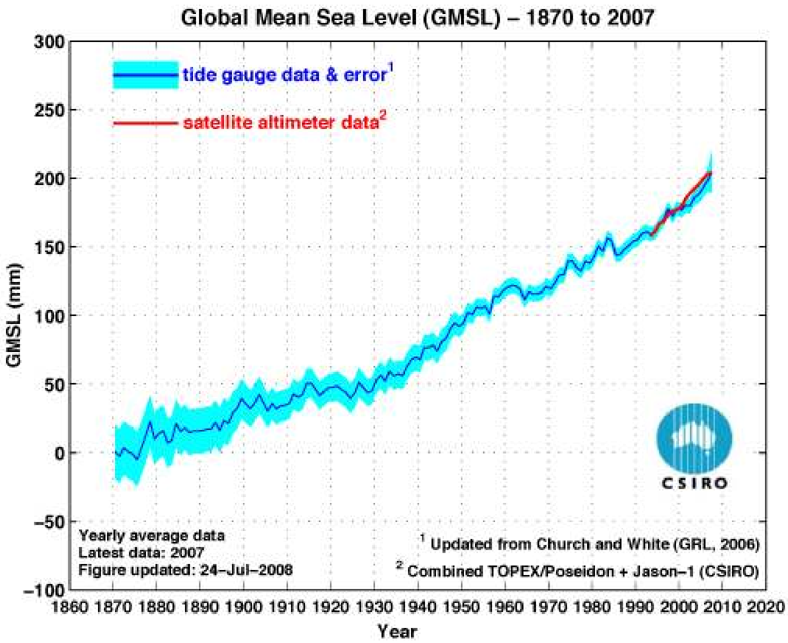
有很多方法可以说明，过度依赖经验会导致对气候变化的困惑或怀疑。我们可以从《澳大利亚人报》(The Australian newspaper)援引 80 岁经常去海滩的凯文·考特(Kevin Court)关于气候变化和海平面上升的声明中看到一个例子:

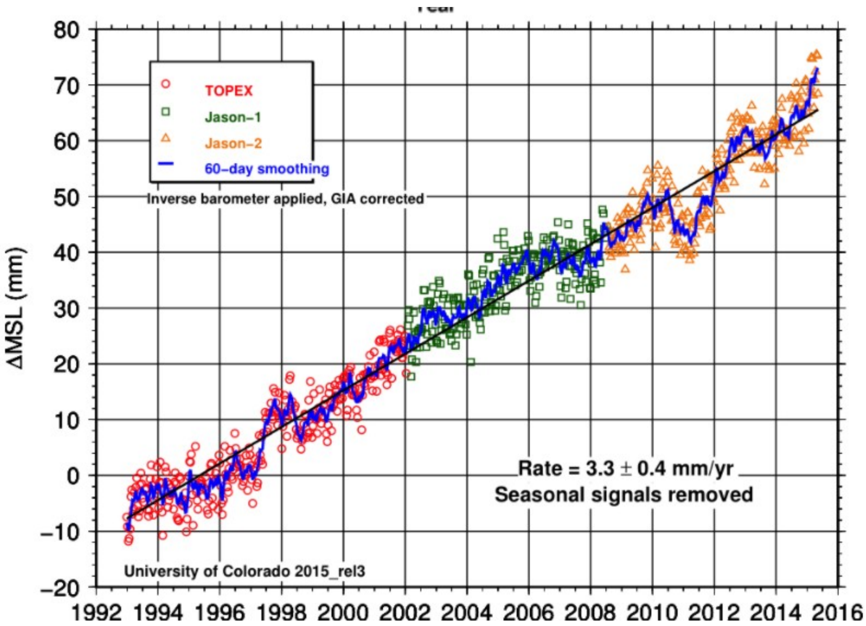
“过去 50 年里，我每天都在这个海滩游泳，这里没什么变化，”考特昨天说，他从伍伦贡北部海滩的海浪中走出来，那里距离肯布拉港的测量站只有短短的划桨距离。

“所有这些关于海平面上升的讨论——我们大多数老年人没有看到任何变化，几十年来我们一直在下降。”

从凯文的角度来看，在过去的几十年里，海平面似乎没有上升，他一直去海滩。自然，他并不关心气候变化对海平面上升的影响，因为他没有察觉到他所习惯的海平面有任何不同。然而，这并不意味着海平面没有上升。

如果我们考虑个人经验之外的其他数据来源，我们可以看到真实的故事可能是完全不同的。下面的图表说明了这一点，显示了自 19 世纪后期以来海平面一直在上升。





Sea level rise from tide gauge and satellite data since 1870 (left) and 1993 (right).

自 1870 年(左)和 1993 年(右)以来，潮汐测量仪和卫星数据显示海平面上升。

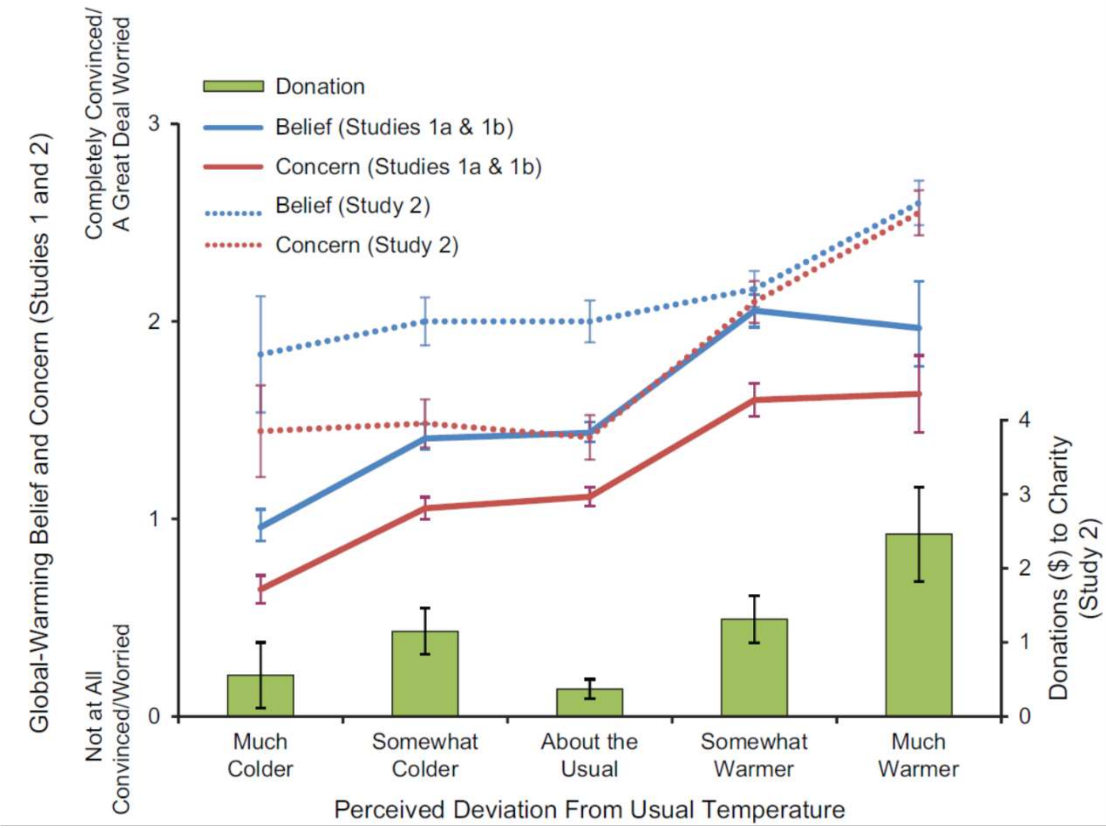
The point is that just because we don't perceive any changes based on our experiences, this doesn't mean that changes aren't actually occurring.

Weather Experiences and Belief in Climate Change

The problem with relying on our experiences becomes even more problematic when we consider the way the weather can influence our strength of belief in climate change. In a study by Li *et al.* (2011), the authors asked people about their belief and concern in climate change alongside how much money they were willing to donate to an environmental charity. In addition to asking these questions, the authors also asked people about their perception of the temperature on the day they did the study. The results are shown in the graph below (Li *et al.*, 2011):

天气经验和对气候变化的信仰

当我们考虑到天气会如何影响我们对气候变化的信念时，依赖我们的经验就变得更加困难了。在李等人(2011)的一项研究中，作者询问了人们对气候变化的信仰和担忧，以及他们愿意为环保慈善机构捐多少钱。除了问这些问题，作者还询问了人们在进行研究当天对温度的感知。结果如下图所示(Li et al.， 2011):



While the graph is complicated, the main points you should take from this are:

1. When people perceive the weather as much **colder** than usual they are: less likely to believe in and be concerned about climate change and are also less likely to donate to climate change charities.
2. Whereas when people perceive the weather as much **warmer** than usual the reverse is true.

This data shows that people will vary in their belief and concern of climate change based on their current experience of temperature. In other words, our concern and belief in climate can fluctuate on a day-to-day basis depending on whether we perceive the temperature as being hotter or colder than average. The problem here is that day to day temperature fluctuations aren't good indicators of long term climate trends. In addition, climate change is also a problem that requires consistent engagement and as this data shows, even minor fluctuations in weather can impact people's willingness to engage with the problem. If you'd like to read more about this, this paper explores additional reasons why warm days change our belief in global warming.

虽然这张图很复杂，但你应该从中得到的主要观点是:

当人们觉得天气比平时冷得多时，他们就不太可能相信和关心气候变化，也不太可能为气候变化慈善机构捐款。

然而，当人们认为天气比平时暖和得多时，情况就相反了。

这一数据表明，人们会根据目前的气温经验，对气候变化有不同的信仰和关注。换句话说，我们对气候的关注和信念每天都在波动，这取决于我们认为气温比平均水平更高或更低。这里的问题是，每天的温度波动并不是长期气候趋势的良好指标。此外，气候变化也是一个需要持续

参与的问题，正如这一数据显示的那样，即使是天气的微小波动也会影响人们参与这个问题的意愿。如果你想了解更多这方面的信息，这篇文章还探讨了为什么温暖的天气会改变我们对全球变暖的看法的其他原因。

习题

The components of psychological distance are:

- Hypothetical, temporal, geographical and interpersonal
- Hypothetical, temporal, spatial and social
- Hypothetical, temporal, spatial and cultural

Under the risk perception model, cognitive factors refers to:

- Our knowledge about the causes of climate change
- Our feelings about climate change
- Our personal experiences with climate change

According to the risk perception model, what is most likely to influence our risk perception of climate change?

- Experiential processing, including emotion and affect.
- Socio-cultural influences, including social norms and socio-cultural influences.
- Cognitive factors, including knowledge.

In a 2011 study by Li *et al*, they found that people:

- Were less concerned about global warming on days they perceived were hotter than average
- Were more concerned about global warming on days that were hotter than average
- Were more concerned about global warming on days when they perceived the temperature as hotter than average

Why do people disagree about the need to act on climate change?

Introduction

In the previous lesson, we looked at how people perceived climate change and the influence of a number of factors such as Psychological Distance and weather experiences on people's perceptions. In this lesson, we will look at what the public think about climate change, as well as the concept of Seepage. We will begin by looking at climate change denial and the seepage effect.

为什么人们在应对气候变化的必要性上存在分歧？

介绍

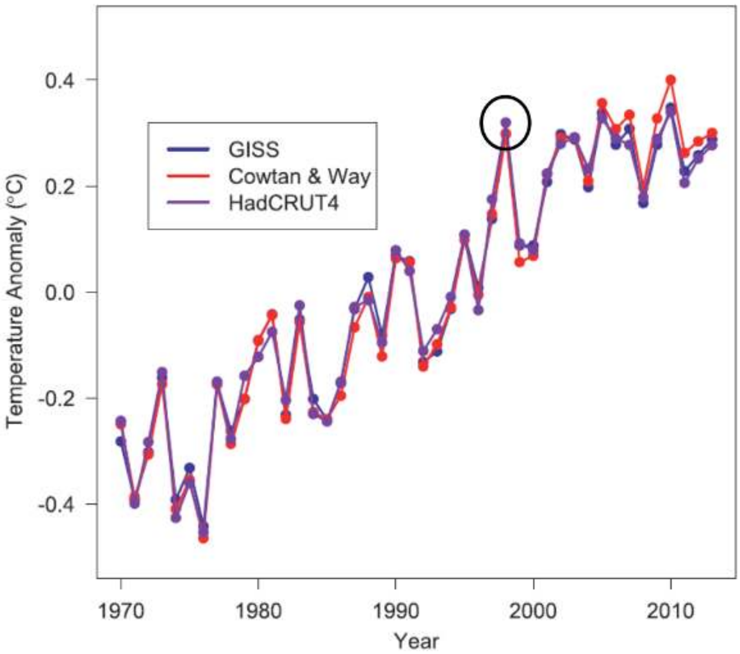
在上节课中，我们了解了人们是如何感知气候变化的，以及心理距离和天气体验等诸多因素对人们感知的影响。在这节课中，我们将了解公众对气候变化的看法，以及渗漏的概念。我们将从气候变化否认和渗透效应开始。

'Seepage' in climate research

One of the things that came up in the video was the contention that climate change has "paused". This argument was very popular during the early 2010's following a particularly warm year in 1998. Temperatures over the next ~15 years seemed to be increasing not nearly as quickly as climate models had predicted. This caused many climate change deniers to argue that climate change had paused or stopped (see the figure below). Unfortunately, this message was adopted by the media and was the subject of several news stories at the time (here are four examples). To rebut this contrarian meme, climate scientists conducted an extensive amount of research investigating the "pause".

气候研究中的"渗透"

视频中提到的一件事是，气候变化已经"暂停"。在经历了 1998 年特别温暖的一年之后，这一观点在 2010 年初非常流行。在接下来的 15 年里，气温似乎没有气候模型预测的那么快。这使得许多否认气候变化的人认为气候变化已经停止或停止(见下图)。不幸的是，这条信息被媒体采纳，并成为当时几则新闻报道的主题(这里有四个例子)。为了反驳这种逆向文化，气候科学家进行了大量的研究，调查"停顿"。

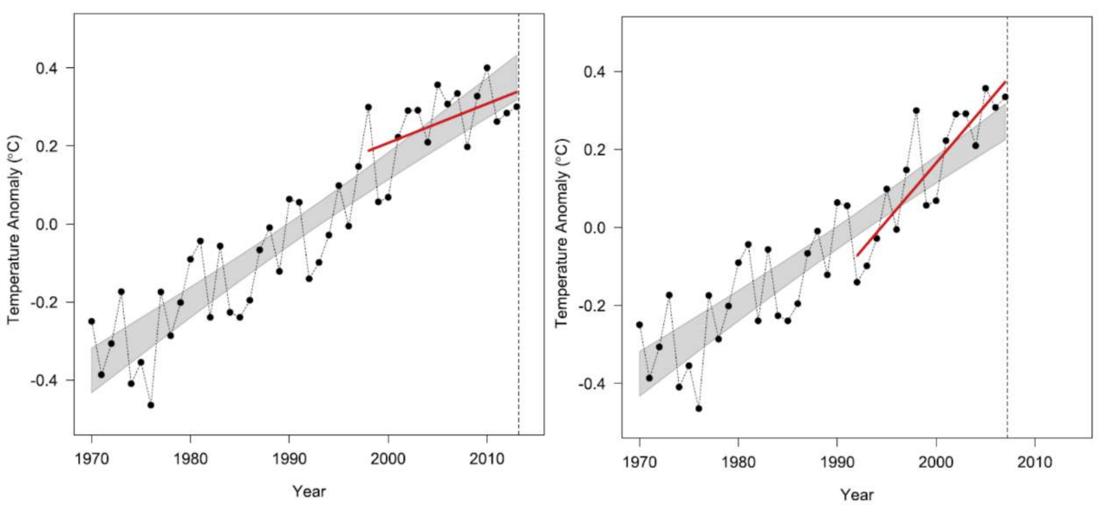


This figure shows global mean surface air temperature anomalies relative to the period 1970-2013 (adapted from Lewandowsky *et al*. 2015). Note 1998 has been highlighted. If you were to ignore years prior to this, it would appear as if very little warming has occurred in the subsequent 15 years.

该图显示了 1970 年至 2013 年期间的全球平均地表气温异常(改编自 Lewandowsky 等人 2015 年的数据)。注 1998 已突出显示。如果你忽略之前的几年，似乎在接下来的 15 年里没有发生什么变暖。

Yet given that the argument of a pause was only based on a very small set of data (climate is usually defined as an average over 30 years), there was insufficient evidence to support the idea that global warming was pausing or slowing down. If we look at the figure below on the left, it shows this apparent 'slow down' between 1998 and 2013, however if we simply take a different 16 year period from 1992 - 2007 (figure below right) we can now make it appear as if warming has accelerated markedly. Working climate scientists were aware that a decrease in the rate of warming on annual to decadal time scales was normal - just natural variability in the climate system moving heat around between the atmosphere and ocean.

然而，考虑到"气候暂停"的论点仅基于一组非常小的数据(气候通常被定义为 30 年的平均水平)，没有足够的证据支持全球变暖正在暂停或减缓的观点。如果我们看看下面左边的图表，它显示了 1998 年到 2013 年之间明显的"减缓"，但是如果我们简单地从 1992 年到 2007 年的 16 年(下图右下方)，我们现在可以让它看起来似乎变暖已经显著加速。从事气候工作的科学家们意识到，每年到十年的变暖速度下降是正常的——只是气候系统在大气和海洋之间传递热量的自然变化。



Left panel (red line) shows linear trend from 1998-2013 while right panel shows linear trend from 1992-2007. Figures adapted from Lewandowsky *et al*. 2015.

左面板(红线)为 1998-2013 年的线性趋势，右面板为 1992-2007 年的线性趋势。数据改编自 Lewandowsky 等人，2015 年。

This meant that the claims of a "pause" were premature and based on a very small amount of data. So why did climate scientists devote so much time, research and funding in refuting the claim of a pause? Perhaps because they felt they needed to address the dissenting voices from the denialist camp and the media. This idea that scientists devote their time and effort to address assumptions or arguments from outside the scientific community brings us to the idea of seepage. We might define it in the following way:

> "[when the] scientific community has adopted assumptions or language from discourse that originated outside the scientific community or from a small set of dissenting scientific voices . . . [and] that those assumptions

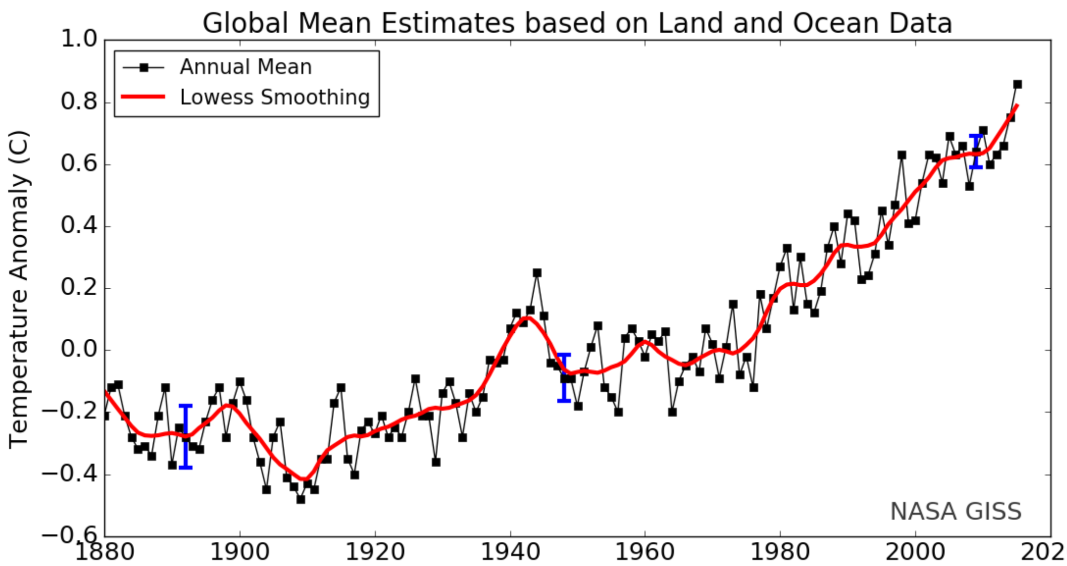
depart from those commonly held by the community."
- **[Lewandowsky *et al*. 2015]**

In other words, seepage is the process of ideas that originate from outside the scientific community affecting how scientists do their work. The problem with seepage is that it can skew scientific inquiry but more importantly, it can also influence the public's perceptions of an issue such as climate change. Ultimately, if so many scientists are doing research investigating an issue, maybe there is some validity in the claim that climate change has paused? Fortunately, most of the community has long since moved on from the idea of a pause, especially considering more recent data such as the figure Brian Cox held up in the debate (see below).

这意味着“暂停”的说法是不成熟的，而且只基于非常少量的数据。那么，为什么气候科学家要花这么多时间、研究和资金来驳斥“暂停”的说法呢?也许是因为他们觉得有必要回应来自否定派阵营和媒体的异议。科学家花费时间和精力来解决来自科学界之外的假设或争论，这种想法给我们带来了“渗透”的想法。我们可以这样定义它:

“(当)科学界采用了来自科学界以外的话语或来自一小群持不同意见的科学声音的假设或语言……[而且]这些假设与社区普遍持有的看法有所不同。”- Lewandowsky 等人 2015

换句话说，渗透是来自科学界之外影响科学家工作方式的想法的过程。渗漏的问题在于，它会歪曲科学探究，但更重要的是，它还会影响公众对气候变化等问题的看法。最后，如果这么多科学家都在做研究调查一个问题，那么气候变化已经停止的说法也许有一定的有效性?幸运的是，大多数社区早已放弃了暂停的想法，特别是考虑到最近的数据，如 Brian Cox 在辩论中提出的数字(见下文)。



Global mean surface air temperature anomalies from 1880-2016 plotted against a 1951-1980 baseline. Red line represents a 5-year smoothing. Image courtesy of NASA.

从 1880-2016 年的全球平均地表气温异常与 1951-1980 年的基线绘制。红线表示 5 年平滑。图片由美国宇航局提供。

While that was a lot of information to take in, we're not expecting you to remember all of it. Instead, the key point you should take away from this section is that the public can influence the focus and way science is conducted. In addition, it's also important that you understand what seepage is and how it has played a role and may continue to play a role in the global warming debate. If you'd like another example of seepage, consider the several papers published on the 'consensus' on climate change. This is another area of research that would not exist were it not for dissenting external voices throwing doubt on the evidence of climate change.

虽然这是很多信息要吸收，我们不期望你记住所有。相反，你应该从这部分中吸取的关键点是，公众可以影响科学的焦点和进行科学的方式。此外，了解什么是渗漏，以及它是如何在全球变暖的争论中发挥作用的也是很重要的。如果你想了解另一个关于渗漏的例子，请考虑发表在气候变化"共识"上的几篇论文。如果没有外部反对声音对气候变化的证据提出质疑，这是另一个研究领域。

Opinions about climate change

For arguments of a pause or for general scepticism about climate change to gain traction, there needs to be some sort of belief that others are of the same opinion. This means that for people like Senator Roberts to gain influence, there has to be some sense that people agree with his sentiments and that there are two sides to the climate debate. It is for this reason that we need to understand what other people think of climate change and how this may influence perceptions of the climate debate.

In a simple study, Leviston *et al.* (2012) gave a large community sample a survey to assess people's responses to this passage: "**Which of the following statements best describes your thoughts on climate change**". Participants were then given four statements and selected the one they most agreed with. The statements are outlined here:

- I don't think that climate change is happening
- I have no idea whether climate change is happening or not
- I think that climate change is happening, but is just a natural fluctuation in Earth's temperatures
- I think that climate change is happening, and I think that humans are largely causing it

The percentage of responses to each statement are shown below:

对气候变化的看法

要想让关于暂停气候变化的争论或对气候变化普遍持怀疑态度的争论获得支持，就需要有某种信念，让其他人持有相同的观点。这意味着，像罗伯茨参议员这样的人要想获得影响力，必须有某种感觉，人们同意他的观点，而且在气候辩论中有两方。正是出于这个原因，我们需要了解其他人对气候变化的看法，以及这可能如何影响人们对气候辩论的看法。

在一项简单的研究中，Leviston 等人(2012)对一个大型社区样本进行了一项调查，以评估人们对这篇文章的反应:"以下哪个陈述最能描述你对气候变化的想法"。然后，参与者被给予四种陈述，并选择他们最同意的一种。这些声明概述如下:

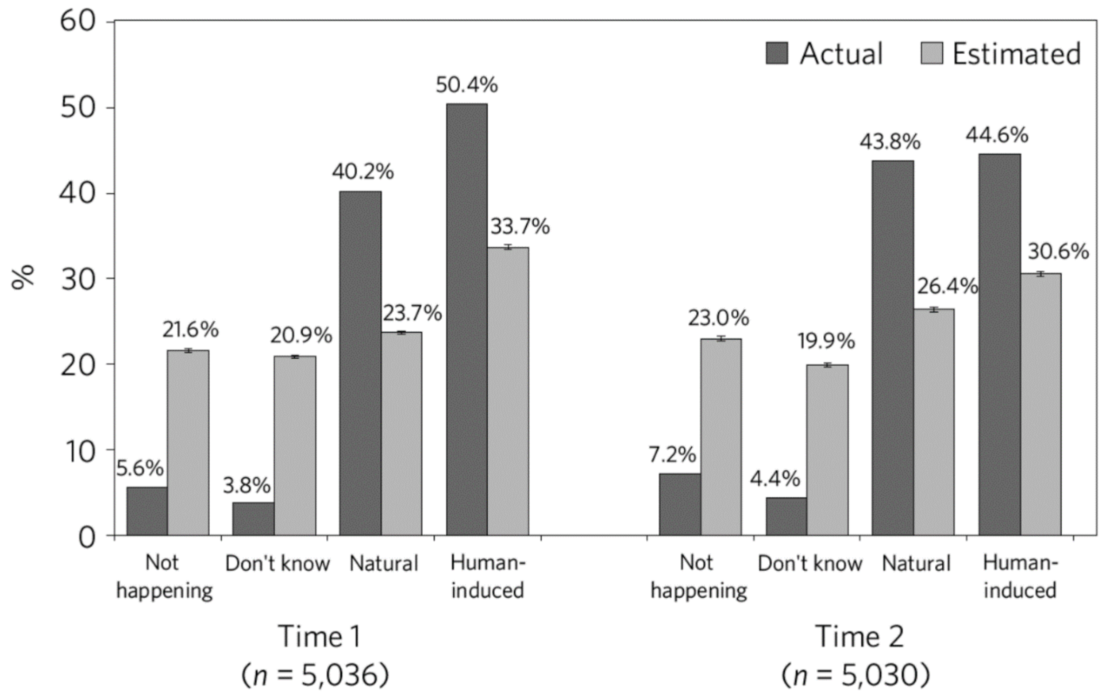
我不认为气候变化正在发生

我不知道气候变化是否正在发生

我认为气候变化正在发生，但只是地球温度的自然波动

我认为气候变化正在发生，我认为人类是主要原因

对每项陈述的答复百分比如下:



Actual vs. Estimated opinion on climate change (Leviston *et al*., 2012). Time 1 and 2 represent the same experiment conducted at two different times (while there are differences, the results are largely the same)

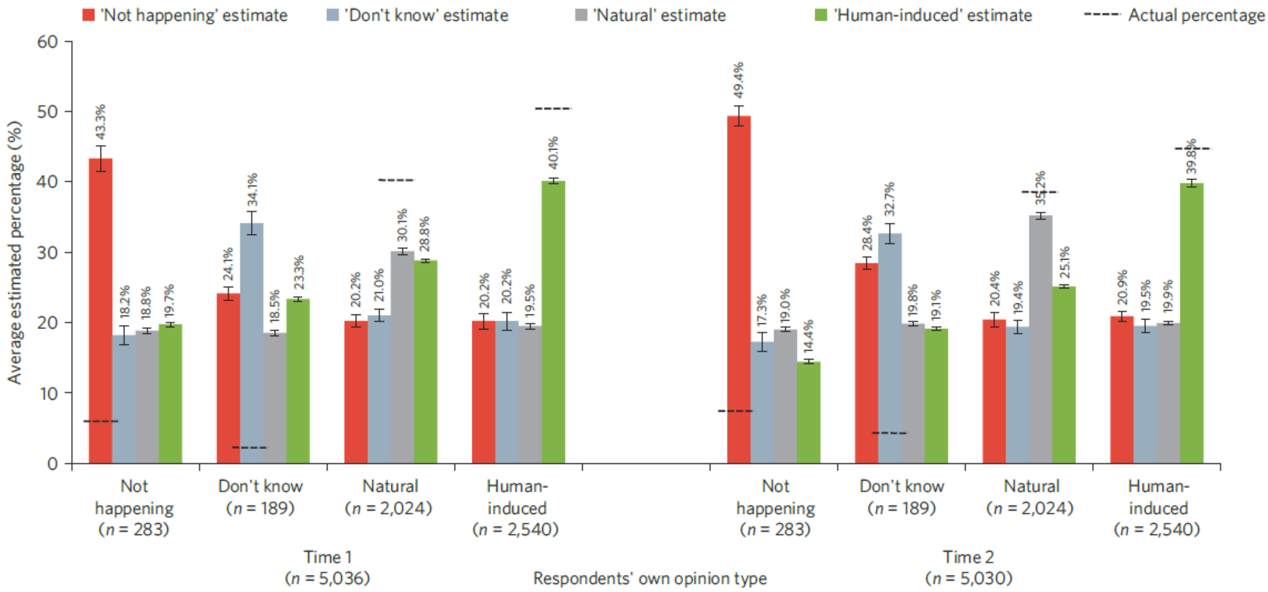
对气候变化的实际意见和估计意见(Leviston 等，2012)。时间 1 和时间 2 代表在两个不同时间进行的相同实验(虽然有差异，但结果基本相同)

The first thing to focus on in this graph is the dark grey bars. These show the number of people who actually reported the viewpoint associated with the question on the horizontal axis. So, this means that at Time 1 (the first of two surveys) only 5.6% of people selected "**I don't think climate change is happening**" as shown in the "**not happening**" column. On the other hand, 50.4% of people surveyed answered "**I think climate change is happening, and I think that humans are largely causing it**" as shown in the "**Human-induced**" column.

In the same study, participants were asked to estimate how they thought other people would respond to the passage: "**Which of the following statements best describes your thoughts on climate change**". The data is shown in the graph below:

在这张图中首先要关注的是深灰色的条形图。横轴上显示了实际报告与问题相关观点的人数。所以，这意味着在第一阶段(两次调查中的第一次)，只有 5.6%的人选择了“我不认为气候变化正在发生”，如“不发生”一栏所示。另一方面，50.4%的受访者回答“我认为气候变化正在发生，我认为人类是主要原因”，如“人为因素”一栏所示。

在同一项研究中，参与者被要求估计他们认为其他人会如何回应这篇文章:“以下哪项陈述最能描述你对气候变化的看法?”数据如下图所示:



This graph may be slightly confusing so lets break it down. The first thing to note is the type of responder on the X axis (the horizontal axis). Here you can see four groups "**Not happening**", "**don't know**", "**natural**" and "**human induced**" for both times the experiments were run (lets just focus on **Time 1** to make it easy). If we look at the "**Not happening**" label on the x-axis, it has four bars representing how people who answered "**I don't think climate change is happening**" think other participants answered the same question. In this case, the people who think climate change is not happening believe that 43% of participants believe the same thing. In reality however, only 5.6% answered this (the dotted horizontal line). Takeaway message? Participants were bad at estimating how many people agreed with their position.

While the "**not happening**" and "**don't know**" respondents greatly overestimated the number of people that agreed with them (again, compare size of bar to where the dotted line is), the people who answered "current climate change is natural" and "current climate change is human induced" both underestimated the number of people who agreed with them.

The "**not happening**" and "**don't know**" groups are displaying a cognitive bias known as the False consensus effect. This bias reflects the overestimation among people that their views or opinions are more widely held or more mainstream than they are in reality. This is why they greatly overestimate the number of people who hold their fringe view that climate change is not occurring.

The second interesting result highlights a phenomenon known as pluralistic ignorance. This effect refers to when someone in a group holds an idea contrary to the norm but believes that most other people believe the norm. In this instance, the 'norm' is a belief that there is a large part of the community that rejects climate science. No matter what the participants answered, they all over-estimated the number of people who rejected the science around climate change (for example, the people who answered that current climate change is human induced overestimated the number of people who think it's not happening - 20.2% as opposed to 5.6% in actuality).

这张图可能有点让人困惑，让我们来分解一下。首先要注意的是 X 轴(水平轴)上的响应器类型。在这里你可以看到四组"不发生"，"不知道"，"自然"和"人为诱导"的实验进行了两次(让我们专注于时间 1)。如果我们看一下 x 轴上的"不发生"标签，它有四个条形图，表示那些回答"我认为气

候变化没有发生”的人认为其他参与者回答了同样的问题。在这种情况下，认为气候变化没有发生的人相信 43%的参与者相信同样的事情。但实际上，只有 5.6%的人回答了这个问题(虚线)。外卖口信吗?参与者不善于估计有多少人同意他们的观点。

而“不发生”和“不知道”受访者的人数大大高估了,同意他们(再一次,比较大小条虚线的地方),那些回答“当前气候变化是自然”和“当前气候变化是人类诱导”都低估了的人同意他们。

“不发生”和“不知道”的群体表现出一种认知偏差，被称为“错误共识效应”。这种偏见反映了人们高估了自己的观点或观点比现实中更被广泛接受或更主流。这就是为什么他们大大高估了持极端观点认为气候变化没有发生的人数。

第二个有趣的结果强调了一种被称为“多元无知”的现象。这种效应指的是当一个群体中的某个人持有与常规相反的想法，但却认为大多数其他人都是这样认为的。在这种情况下，“规范”是一种信念，即社区中有很大一部分人拒绝接受气候科学。无论参与者回答什么,他们都高估了的人拒绝了科学在气候变化(例如,回答的人,当前气候变化是人为高估的人认为这是没有发生现状-20.2%与 5.6%)。

Why are these biases happening?

So, what might be causing these effects? One explanation is that phenomenon such as the false consensus effect are caused as a result of confirmation bias; that is, the tendency to search for evidence consistent with our views and to ignore evidence that is inconsistent with our views. In the case of the climate change debate, people may deny climate change because they ignore evidence supporting climate change and seek out evidence against climate change. In addition, people also have a natural tendency to associate with those who share similar political and world views. However, if the only people you associate with share your political views, you may come to believe that these political views are commonly shared by everyone else in the community.

In the case of pluralistic ignorance, the media may play a significant role. For instance, news coverage of climate change over the past decade was often portrayed as if there were still significant uncertainty or debate in the scientific community when it comes to our confidence in the science. This can encourage a false perception that there are many more people out there who don't believe climate change is occurring than exist in reality. If you haven't watched it already, John Oliver demonstrates this well on his show.

为什么会出现这些偏见?

那么，是什么导致了这些影响呢?一种解释是，错误共识效应等现象是由确认偏误引起的;也就是说，人们倾向于寻找与自己观点一致的证据，而忽视与自己观点不一致的证据。在气候变化的辩论中，人们可能会否认气候变化，因为他们忽视了支持气候变化的证据，并寻求反对气候变化的证据。此外，人们也有一种自然的倾向，与那些有相似的政治和世界观的人交往。然而，如果你联系的唯一的人与你有相同的政治观点，你可能会认为这些政治观点是社区中其他人共同拥有的。

在多元无知的情况下，媒体可能发挥了重要作用。例如，过去十年里关于气候变化的新闻报道常常被描述成科学界在我们对科学的信心方面仍然存在重大的不确定性或争论。这可能会助长一种错误的看法，认为不相信气候变化正在发生的人比现实中存在的人要多。如果你还没看过，约翰·奥利弗在他的节目中很好地展示了这一点。

Despite the fact that 97% of published climate science agrees that global warming is happening and being caused by humans, only about half of the population agree. So why isn't the message getting through to people? There are a few reasons and we'll touch on some of them briefly here. The first reason is that people may have their heads in the sand, that is, psychological distance may be playing a big role influencing peoples' belief in and concern for, climate change (as discussed in lesson 1). The second reason could be that people don't have sufficient scientific literacy skills to understand climate change. The final reason could be that people are politically motivated to deny climate change. We will take a closer look at some of these ideas in the next lesson.

尽管 97%已发表的气候科学都认为全球变暖正在发生，而且是由人类造成的，但只有大约一半的人同意这一观点。那么，为什么人们不能理解这一信息呢?有几个原因，我们将在这里简要地谈到其中的一些。第一个原因是,人们可能有他们的头埋在沙子里,也就是说,心理距离可能会发挥巨大的作用影响着人们的信仰和关心,气候变化(如第一课中讨论)。第二个原因可能是,人们没有足够的科学素养能力理解气候变化。最后一个原因可能是人们出于政治动机而否认气候变化。在下一课中，我们将仔细研究这些观点。

习题

The seepage effect refers to the idea that:

- Scientists can be influenced by information/ opinions from outside the scientific community

- People are influenced by one another as a result of public discourse

- The idea that everyone overestimates the amount of climate change deniers

In a survey of people's opinions on climate change, Leviston *et al*. (2012) found that:

- Most people think their opinions are the least common
- Most people think that there are more climate change deniers than there actually are
- Most people answer "don't know" if asked whether climate change is happening

One explanation for the false consensus effect is that:

- People often selectively search for information consistent with their world views while discounting evidence inconsistent with these views

- People don't like to think they're wrong

Scientific literacy

A common claim made by many, is that people aren't engaged with climate change because they don't have enough scientific literacy. The idea here is that if people aren't educated enough about science then maybe this is why people deny or ignore the need to act against climate change. On the flip side, those with higher science literacy should be more accepting of climate change because they have greater knowledge of the scientific process. This argument is known as the science comprehension thesis (SCT).

A study by Kahan et al. (2012) tested this thesis through the use of a survey in which they asked 1540 US Citizens the following question: "How much risk do you believe climate change poses to human health, safety or prosperity?". They then measured people's science literacy and looked at whether there was a relationship between people's scientific literacy and their ratings of climate change risk.

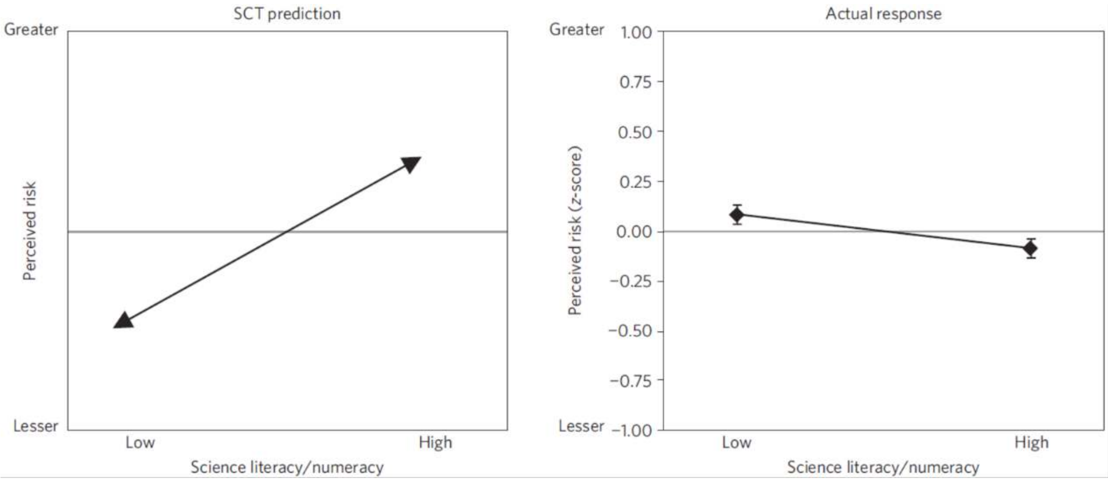
The graphs below depict the results of this study. On the left, we have the predicted relationship between climate change risk perceptions and science literacy as under the SCT hypothesis. The graph on the right depicts the actual data from the study.

科学素养

许多人普遍认为，人们之所以不参与气候变化，是因为他们没有足够的科学素养。这里的观点是，如果人们没有接受足够的科学教育，那么这可能就是人们否认或忽视应对气候变化的必要性的原因。另一方面，那些科学素养较高的人应该更容易接受气候变化，因为他们对科学过程有更多的了解。这一论点被称为科学理解命题(SCT)。

Kahan 等人(2012)的一项研究对这篇论文进行了测试，他们在调查中向 1540 名美国公民提出了以下问题:“你认为气候变化对人类健康、安全或繁荣造成多大风险?”然后，他们测量了人们的科学素养，并观察人们的科学素养和他们对气候变化风险的评级之间是否存在关联。

下面的图表描述了这项研究的结果。在左边，我们有在 SCT 假设下气候变化风险感知和科学素养之间的预测关系。右边的图表描述了这项研究的实际数据。



Predicted vs. actual perceived risk of climate change plotted against scientific literacy. Figure adapted from [Kahan *et al*. 2012].

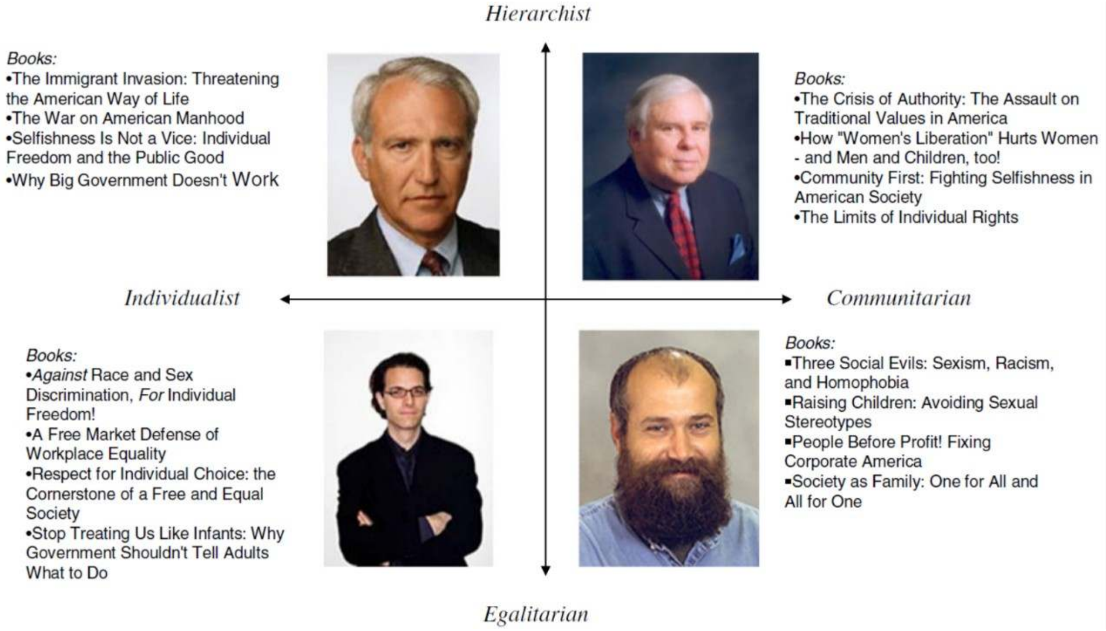
预测和实际感知的气候变化风险与科学素养之间的关系。图改编自 Kahan 等人，2012。

As you can see, the results from the study contradict the scientific comprehension thesis as they show those with higher scientific literacy actually perceive climate change as a lower risk, whereas those with lower scientific literacy perceive climate change as a greater risk.

Kahan and colleagues (2012) didn't stop there. Instead, they looked further into the data by analysing the role of someone's world view on perceived climate change risk. By using a political questionnaire, the researchers categorised responders based on their identity under two world view dimensions: **hierarchical/egalitarian** identity and **individualist/communitarian** identity. The figure below loosely depicts people based on these identities. In their analysis, Kahan *et al*. (2012) chose to focus on hierarchical/individualists and communitarian/egalitarians. That is, the types of individuals depicted in the upper left and lower right quadrants.

正如你所看到的，这项研究的结果与科学理解的论点相矛盾，**因为它们表明，那些具有较高科学素养的人实际上认为气候变化的风险更低，而那些具有较低科学素养的人认为气候变化的风险更大。**

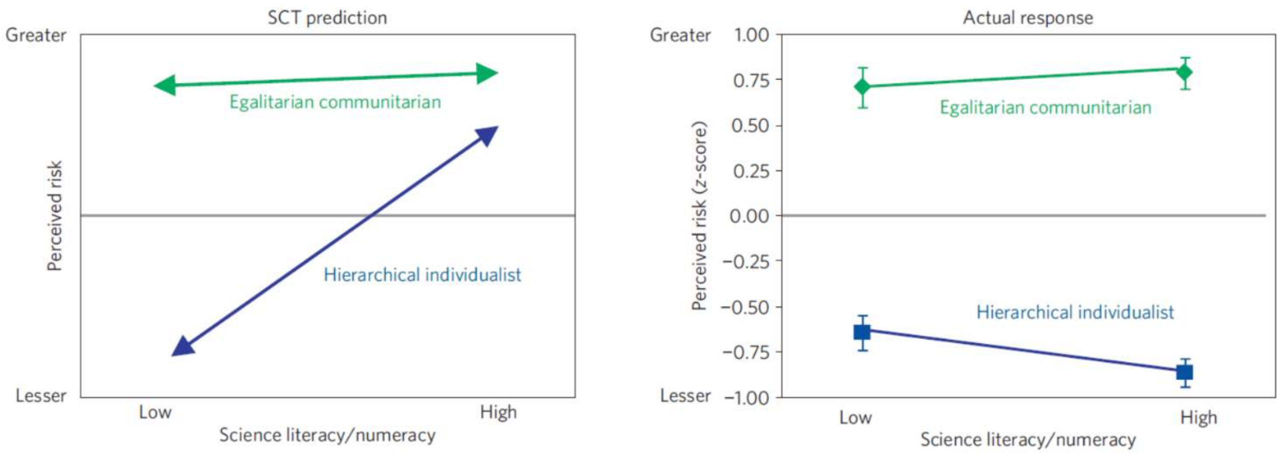
卡汉和他的同事(2012 年)并没有就此止步。相反，他们通过分析某人的世界观在感知气候变化风险中所起的作用，进一步研究了这些数据。通过政治问卷，研究人员根据参与者的身份在两个世界观维度下进行分类:等级/平等主义身份和个人/社群主义身份。下图基于这些身份粗略地描述了人们。在他们的分析中，Kahan et al.(2012)选择了关注等级/个人主义者和社群/平等主义者。也就是说，在左上和右下象限中描绘的个体类型。



Different (fictional) people who represent the four quadrants of the socio-political spectrum (Kahan *et al.* 2010).

Hierarchist 阶级主义 Egalitarian 平等主义 Individualist 个人主义 Communitarian 共产主义

The graphs below show the predictions of the SCT hypothesis alongside the actual data from the study.



The same results as above, except respondents are broken down into two different socio-political groups. Figure adapted from Kahan *et al.* 2012.

结果与上述相同，只是受访者被分为两个不同的社会政治群体。图改编自 Kahan 等人，2012。

As you can see, an increase in scientific literacy makes little if any difference for the "egalitarian communitarian" group - they perceive climate change as a risk regardless (the green line in the right panel stays high - greater risk perception, but flat - no difference as a function of low or high science literacy). Hierarchical individualists however actually decrease their perceived risk of climate change as their scientific literacy increases (the blue line in the right panel is lower - lower risk perception - and drops further still for those with high science literacy). This data once again, contradicts the scientific comprehension thesis and is summarised nicely by the authors as "Contrary to SCT's predictions, high science literate and numerate hierarchical individualists are more sceptical, not less, of climate change risks." Based on these results, it appears our world view seems to moderate how we perceive scientific issues such as climate change much more than our scientific literacy. Kahan et al. (2010) suggest that "individuals are motivated selectively to credit empirical claims in a manner that coheres with their visions of the ideal society". This is a classic example of motivated cognition.

正如你所看到的,提高科学素养本身没有多少区别的“平等的社群主义”,他们认为气候变化风险不管(绿线在右边面板高企——更大的风险知觉,但平——没有区别低或高科学素养)的函数。然而，等级个人主义者实际上会随着他们科学素养的提高而降低他们所感知到的气候变化风险(右边的蓝线表示较低的风险感知，而对于那些科学素养较高的人来说，风险感知更低)。这一数据再次与科学理解的论点相矛盾，作者将其很好地总结为:“与 SCT 的预测相反，**具有较高科学素养和计算能力的等级个人主义者对气候变化风险的怀疑更多，而不是更少。**”从这些结果来看，我们的世界观似乎比我们的科学素养更能缓和我们对气候变化等科学问题的看法。Kahan 等人(2010)认为，“个人被选择性地激励去相信与他们对理想社会的愿景相一致的经验主义主张”。这是动机认知的一个经典例子。

Scientific consensus

So, if world view influences perception of climate change and science literacy doesn't appear to fix this problem, what can we do about increasing climate change acceptance? One thing we can do is increase people's perception of scientific consensus on climate change.

Recall that in the previous section we discussed how those with hierarchical individualist views were less likely to accept climate change if they had high amounts of scientific literacy. While this can be a problem, a study by Stephan Lewandowsky and colleagues demonstrated that knowledge of the scientific consensus on climate change can increase the likelihood that hierarchical/individualists will accept the science.

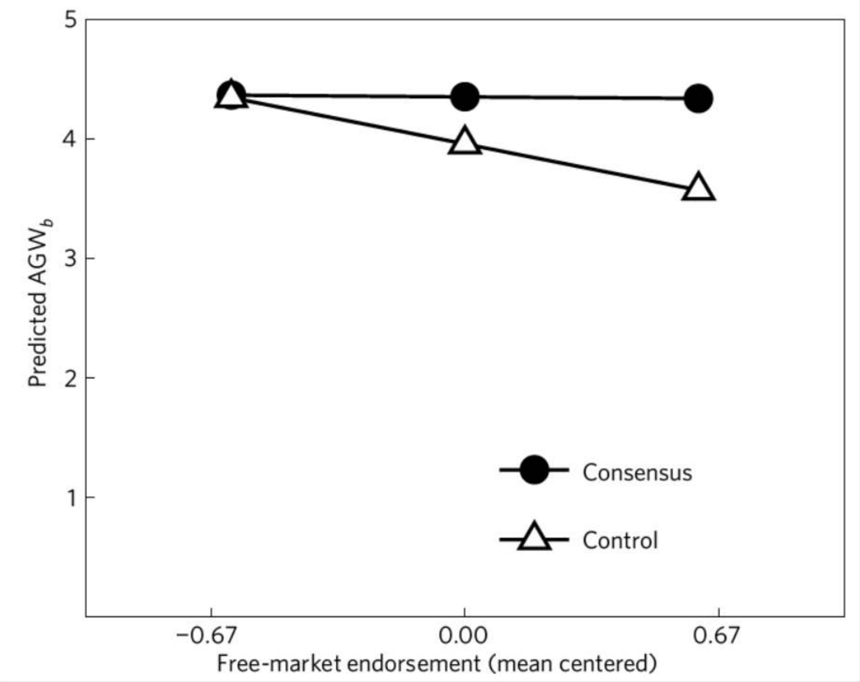
In their study, Lewandowsky *et al*. (2013) gathered a group of participants and separated into two groups; a "consensus" group and a "control group". The consensus group received the following information: "**97 out of 100 experts agree that global warming is a consequence of burning of fossil fuels**". The control group did not receive this information. Following this Lewandowsky and colleagues plotted people's belief in anthropogenic (human caused) global warming (AGW) as a function of their free market endorsement. The results of this study are shown below.

科学共识

那么，如果世界观影响了人们对气候变化的看法，而科学素养似乎不能解决这个问题，我们能做些什么来增加人们对气候变化的接受度呢?我们能做的一件事就是增加人们对气候变化的科学共识的认识。

回想一下，在前面的章节中，**我们讨论了那些持有等级个人主义观点的人，如果他们有很高的科学素养，他们是如何不太可能接受气候变化的。**虽然这可能是个问题，但斯蒂芬·莱万多夫斯基(Stephan Lewandowsky)及其同事的一项研究表明，**对气候变化的科学共识的了解可以增加等级/个人主义者接受科学的可能性。**

在他们的研究中，Lewandowsky 等人(2013)将一组参与者分成两组;一个“共识组”和一个“控制组”。共识组收到的信息如下:“97 / 100 的专家同意全球变暖是燃烧化石燃料的后果”。对照组没有收到这些信息。在此之后，莱万多夫斯基和同事们绘制了人们对人为(人类引起的)全球变暖(AGW)的信念作为自由市场支持的函数。本研究结果如下所示。



Acceptance of AGW vs. Endorsement of the free market. Figure adapted from Lewandowsky *et al*. (2013).

Note that in this graph, people who rank highly on free market endorsement are more likely to be hierarchical individualist while those who rank lowly on free market endorsement are more likely to be egalitarian communitarians. In the graph, we can see that those given information about scientific consensus (consensus group) have higher perceived belief in anthropogenic global warming regardless of their level free-market endorsement. Conversely, those in the control group had less belief in AGW the more they endorsed the free market. So, for hierarchical individualists it seems as if knowledge of the scientific consensus plays the greatest role in influencing whether these individuals accept climate change. But given that 97% of climate scientists are in agreement, why is it that people are still sceptical about climate change in general? Once again, the video by John Oliver captures the problem.

请注意，在这张图表中，在自由市场背书上排名靠前的人更可能是等级个人主义者，而那些在自由市场背书上排名靠后的人更可能是平等的社群主义者。在图表中，我们可以看到，那些给出科学共识信息的人(共识群体)对人为全球变暖有更高的感知信念，不管他们对自由市场的认可程度如何。相反，对照组的人越认同自由市场，对 AGW 的信念就越少。**因此，对于等级个人主义者来说，似乎科学共识的知识在影响这些个人是否接受气候变化方面发挥了最大的作用。**但是既然 97%的气候科学家都同意这一观点，为什么人们仍然对气候变化普遍持怀疑态度呢?约翰·奥利弗(John Oliver)的视频再次抓住了这个问题。

Hopefully you picked up on the issue. If not, one of the reasons people don't accept climate change despite there being overwhelming scientific consensus, is that the media often portray climate change as being:

1. A debate and not a fact
2. A debate between two scientists of differing opinions as opposed to 97 climate scientists who accept climate change contrasted 3 scientists who do not.

While the media is often obliged to present contentious issues in a debate style format, the impression this gives to the public is often flawed. Instead of perceiving the climate change 'debate' as occurring between 97% of climate scientists who accept climate change compared to the 3% who do not, the public may readily perceive the issue as an even debate between equal numbers of people in the climate science community.

This phenomenon can be loosely attributed to the availability heuristic.

希望你注意到了这个问题。如果不是这样，人们不接受气候变化的原因之一是，尽管有压倒性的科学共识，媒体经常描述气候变化为:

只是一场辩论，而不是一个事实

两名持不同观点的科学家与 97 名接受气候变化的科学家之间的辩论，与 3 名不接受气候变化的科学家形成了对比。

虽然媒体经常被迫以辩论形式提出有争议的问题，但这给公众的印象往往是有缺陷的。公众不认为气候变化“辩论”发生在接受气候变化的 97%的气候科学家之间，而认为不接受气候变化的只有 3%，他们可能很容易把这个问题看作是气候科学界同等人数之间的辩论。

这种现象可以松散地归因于可得性启发式。

Understanding uncertainty

In the past few lessons, we've discussed a number of factors influencing people's perception of climate change such as experience, psychological distance and consensus. The last thing we will touch upon is the role that people's understanding of uncertainty plays in their comprehension of climate science.

理解的不确定性

在过去的几节课中，我们讨论了一些影响人们对气候变化感知的因素，如经验、心理距离和共识。我们最后要谈到的是人们对不确定性的理解在他们对气候科学的理解中所起的作用。

If a weather forecast states "There is a 30% chance of rain tomorrow" which of these three statements do you think best characterises the intended meaning:

- (•) It will rain on 30% of the days like tomorrow
- ( ) It will rain tomorrow in 30% of the region
- ( ) It will rain tomorrow 30% of the time

Uncertainty

The correct answer was "It will rain on 30% of the days like tomorrow". If you got the answer wrong don't worry, quite a few people get this question wrong depending on their familiarity with these types of questions and their cultural backgrounds. Nevertheless, this example illustrates one of the common problems people have with probabilistic reasoning alongside their ability to correctly apply probability statements.

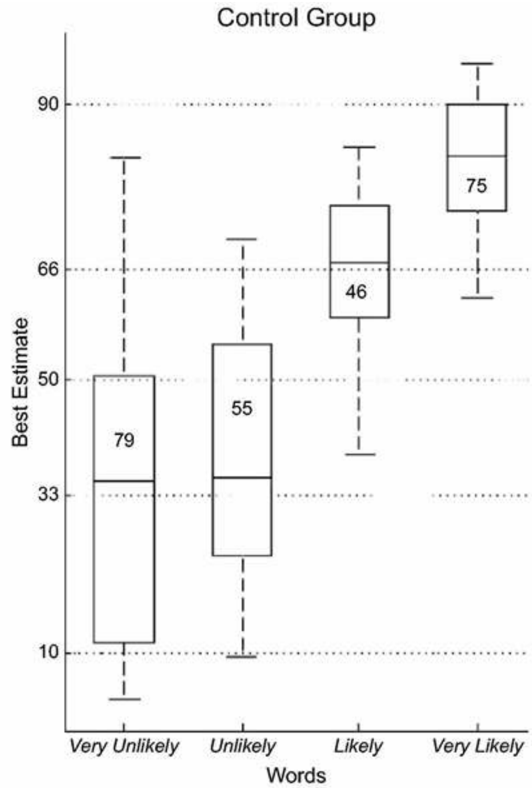
Probabilistic statements are commonly used by the Intergovernmental Panel on Climate Change (IPCC) to convey information about long term climate predictions. However, if people fail to correctly understand probabilistic statements, this can lead to misinterpretations.

In a study looking at peoples' interpretation of probabilistic statements, Budescu *et al.* (2009) asked participants to rate the likelihood they thought corresponded with statements which included a number of descriptive terms such as "very unlikely", "unlikely", "likely" and "very likely". For example, "It is very unlikely that hot extremes, heat waves, and heavy rain will not continue to become more frequent" Participants were then asked to estimate the numerical probabilities that corresponded with these terms.

不确定性
正确答案是"像明天这样 30%的日子会下雨"。如果你答错了，不要担心，很多人会因为对这类问题的熟悉程度和他们的文化背景而做错这道题。然而，这个例子说明了一个常见的问题，人们有概率推理和他们的能力正确应用概率陈述。

政府间气候变化专门委员会(IPCC)通常使用概率性陈述来传达有关长期气候预测的信息。然而，如果人们不能正确理解概率性陈述，就会导致误解。

在一项研究中，Budescu 等人(2009)要求参与者评价他们认为包含"非常不可能"、"不可能"、"可能"和"非常可能"等描述性术语的陈述对应的可能性。例如，"极端高温、热浪和暴雨不可能继续变得更加频繁"，然后要求参与者估计与这些术语相对应的数字概率。



The box and whisker plots show the range of numerical value that participants associated with the terms on the x-axis. The numbers in each box are the percentage of participants that incorrectly guessed the value associated with the term. Figure adapted from Budescu *et al.* (2009).

盒形图和须形图在 x 轴上显示了参与者与这些术语相关联的数值范围。每个方格里的数字是不正确猜出与该术语相关的值的参与者的百分比。图改编自 Budescu 等人(2009)。

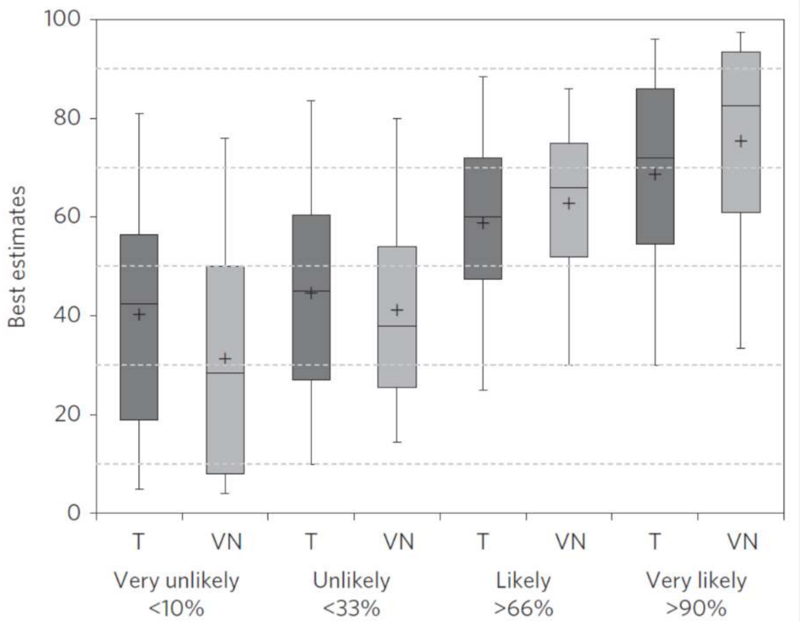
Looking at the graph above, we can see that when people are asked to convert probability statements such as 'very unlikely' into numerical probabilities, 79% of people respond incorrectly. Similarly, statements that include 'very likely' saw 75% of people respond incorrectly. The intended meaning was 'less than 10%' for 'very unlikely' and 'greater than 90%' for 'very likely', but the majority of people over estimated the meaning of 'very unlikely' and under estimated the meaning of 'very likely'. In addition, note the large spread of estimates for each term - 'very unlikely' can mean anything from ~5% to ~80% depending on who you ask. This can be a significant problem when the public interpret probability statements written by the IPCC and climate scientists because they may incorrectly interpret the probability of an event occurring (e.g., global temperature rise or the certainty with which we understand the science).

One thing we can do to improve people's estimates is to give them the intended numerical probabilities in conjunction with each probability statement. For example, "It is very unlikely (<10%) that hot extremes, heat waves, and heavy rain will not continue to become more frequent".

从上面的图表中，我们可以看到，当人们被要求将“非常不可能”等概率表述转换为数字概率时，79%的人的回答是错误的。类似地，如果陈述中包含“很有可能”，75%的人的回答都是错误的。意思是‘less than 10%’表示‘非常不可能’，‘greater than 90%’表示‘very likely’，但是大多数人高估了‘very unlikely’的意思，低估了‘very likely’的意思。此外，请注意每个词的估计相差很大——“非常不可能”可能意味着从 5%到 80%不等，取决于你问的是谁。当公众解读

IPCC 和气候科学家撰写的概率声明时，这可能是一个重大问题，因为他们可能错误地解读某一事件发生的概率(例如，全球气温上升或我们理解科学的确定性)。

我们可以做的一件事是改善人们的估计，即给他们预期的数字概率与每个概率声明。例如，“极端高温、热浪和暴雨不会继续变得更加频繁的可能性很小(<10%)。”



The graph above depicts people's estimation of probability in a follow up study where Budescu and colleagues (2014) provided people with verbal numerical statements (such as the example provided beforehand) or with a translation table that helped to translate written probability statements into numerical probability statements. In the graph, the plus symbol represents the average estimate of participants for the probability statements listed, while the letter "T" refers to the translation group, and the letters "VN" refer to the verbal numerical group. So, when people are asked to convert probability statements such as "very unlikely" into numerical probabilities, the translation group, estimate this probability to be 40%, whereas the verbal numerical group estimate the probability to be 30%. This is in contrast with the intended probability of the statement very unlikely (<10%), but while the verbal numerical group are incorrect with their estimate they still perform better than the translation group.

For reference, here is how the IPCC defines its probability statements:

上图描述了在一项后续研究中人们对概率的估计，Budescu 和同事(2014)向人们提供了口头数字陈述(如之前提供的例子)，或者提供了一个帮助将书面概率陈述转化为数字概率陈述的翻译表。图中，加号代表参与者对所列概率陈述的平均估计，字母“T”代表翻译组，字母“VN”代表言语数字组。所以，当人们被要求将诸如“非常不可能”之类的概率陈述转换为数字概率时，翻译组的人估计这个概率为 40%，而语言数字组的人估计这个概率为 30%。这与“非常不可能”(<10%)的预期概率形成对比，但是，虽然口头数字组的估计是错误的，但他们仍然比翻译组表现得更好。

以下是 IPCC 对其概率声明的定义，供参考:

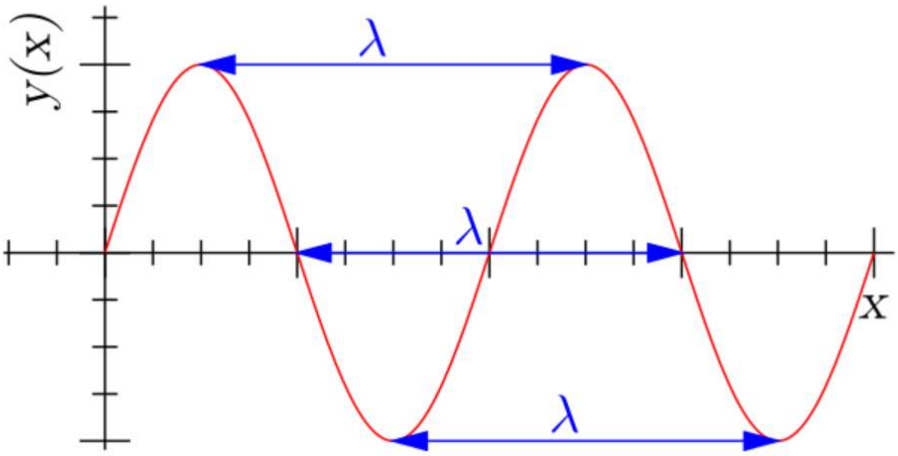
Term	Likelihood of the Outcome
Virtually certain	99–100% probability
Very likely	90–100% probability
Likely	66–100% probability
About as likely as not	33–66% probability
Unlikely	0–33% probability
Very unlikely	0–10% probability
Exceptionally unlikely	0–1% probability

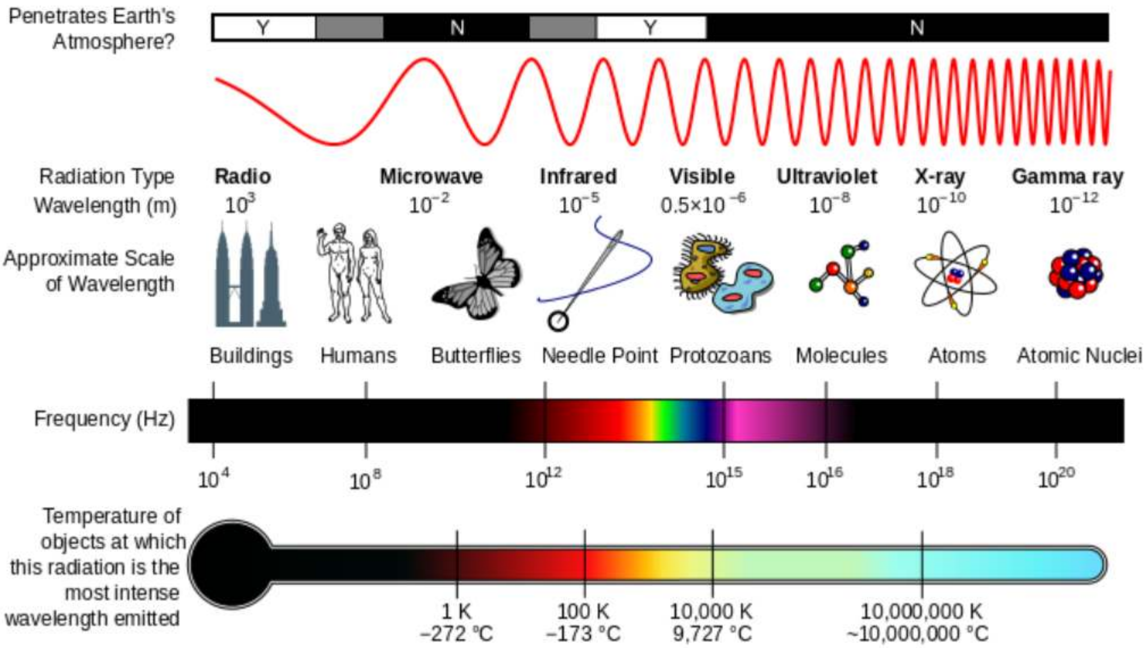
Week2

Week3

Electromagnetic (EM) radiation is a form of energy that is all around us and it takes many forms, such as radio/TV waves, microwaves (like in a microwave oven), X-rays, light, and heat.

The wavelength of EM radiation is defined as the distance between one complete cycle of the wave (See figure below, where wavelength of the red sine wave is given as the Greek letter, lambda - λ).



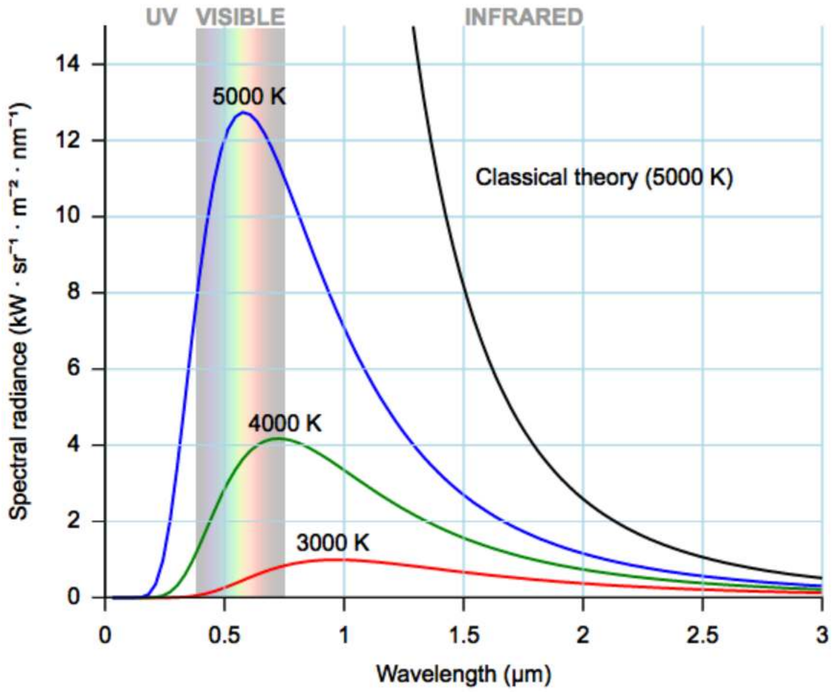


(重点：①物体的温度越高，发出的电磁波频率越高、波长越短。

② 每个波段的频率由低至高依次排列的话，它们是工频电磁波、无线电波（分为长波、中波、短波、微波）、红外线、可见光、紫外线、X 射线及 γ 射线。)

Gamma rays have the smallest wavelengths and the most energy of any wave in the electromagnetic spectrum. They are produced by the hottest and most energetic objects in the universe, such as neutron stars and pulsars, supernova explosions, and regions around black holes.

Any substance with a temperature above absolute zero emits electromagnetic radiation, and it emits this radiation with a range of different wavelengths. The range of EM radiation wavelengths a substance will emit, as well as the intensity of radiation at different wavelengths, is determined by something called a blackbody curve. An example of three different blackbody curves (shown in red, green and blue) is in the figure below. It shows wavelength on the horizontal axis, and the intensity of EM radiation on the vertical axis.



两个定律，不需要记忆公式：

The first is the Stefan-Boltzman law, which simply states that the total energy emitted by an object, across all wavelengths (for each second and square metre of surface area), is proportional to its temperature.

$$E = \sigma T^4$$

Temperature

Energy

Stefan-Boltzmann constant

In this equation, E is the total radiated energy from a blackbody, sigma is the Stefan-Boltzmann constant and T is absolute temperature. From this, it should be clear that **as temperature increases, so does the amount of radiated energy.**

The second is Wien's law which states that the wavelength at which an object's electromagnetic radiation is most intense is inversely proportional to temperature ($\lambda_{max} = b/T$). In simple terms, this means that **as something gets hotter, the peak wavelength of emitted EM radiation will get shorter.**

Wien's constant

$$\lambda_{max} = \frac{b}{T}$$

Peak wavelength

Temperature

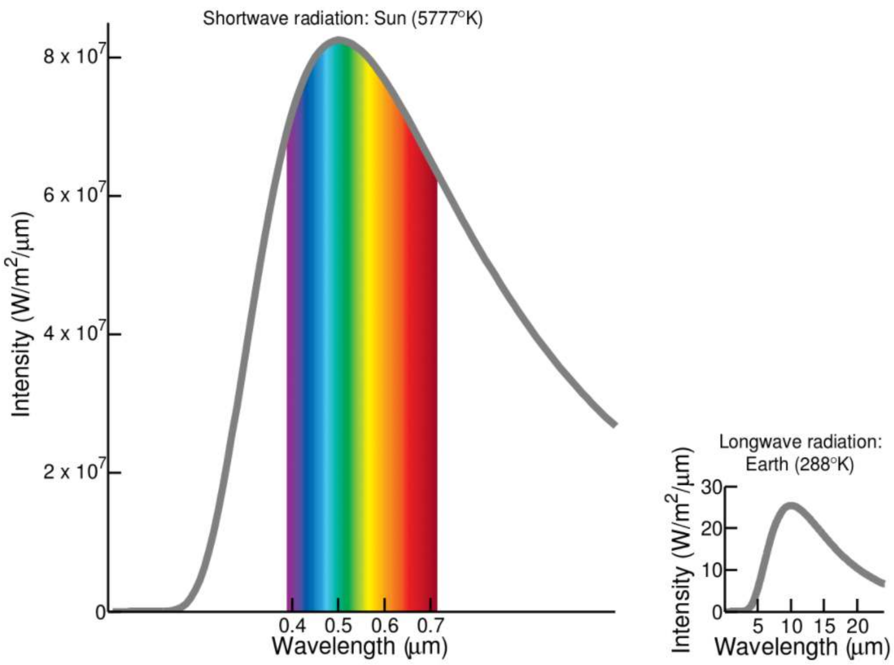
Looking at the figure below, you can see that the sun, with a temperature of around 6000˚ Kelvin emits radiation most strongly in the part of the EM spectrum that we know as visible light - wavelengths around 0.4 -

0.7 micrometres (μm). The Earth meanwhile, with a much lower temperature, emits energy back into space with much longer wavelengths - in the infrared part of the EM spectrum.

看看下面的图，你可以看到温度为 6000˚Kelvin 的太阳发出的电磁光谱中最强烈的辐射，即我们所知的可见光波长，波长约为 0.4 - 0.7 微米(μm)。与此同时，地球的温度要低得多，它向太空释放的能量波长要长得多——在电磁光谱的红外部分。

For this reason, climate scientists often divide the EM spectrum into two parts, at about 1μm. In the Earth system, the higher energy side of this division - "shortwave" radiation - comes from the sun and heats the Earth, and the lower energy side - "longwave" radiation - is what the Earth emits back out to space.

因此，气候科学家经常将电磁光谱分为两部分，在 1μm 左右。在地球系统中，这种划分中能量较高的一面——“短波”辐射——来自太阳并给地球带来热量，而能量较低的一面——“长波”辐射——则是地球释放回太空的。



习题：

1.Which of the following statements is correct?

Visible light is low energy, longwave radiation.

解析：**Visible light** is a form of electromagnetic (EM) radiation, which is visible to most human eyes. Visible light falls in the range of the EM spectrum between infrared (IR) and ultraviolet (UV).（**红外线**（英语：Infrared，简称 IR）是波长介乎微波与可见光之间的电磁波，其波长在 760 奈米（nm）至 1 毫米（mm）之间，是波长比红光长的非可见光，对应频率约是在 430 THz 到 300 GHz 的范围内。**紫外线**（英语：Ultraviolet，简称为 UV），为波长在 10nm 至 400nm 之间的电磁波，波长比可见光短，但比 X 射线长。太阳光中含有部分的紫外线。200 纳米的紫外线辐射会被空气强烈的吸收，因此称之为真空紫外线。）

在地球系统中，太阳的辐射，主要是可见光，代表了比其他电磁辐射更高的能量，更短的波长。这就是为什么气候科学家称其为短波辐射。

As an object gets hotter, it will emit less shortwave radiation and more longwave radiation.
解析：反了

The Earth loses energy by emitting infrared radiation into space.

The Earth is cooler than the sun, so it radiates less longwave radiation into space than it absorbs in shortwave radiation from the sun.
解析：While the Earth is cooler than the sun, this simply means that it emits less energy into space than the sun does. Our planet is (generally speaking) in radiative equilibrium, which means that the energy Earth radiates into space must equal the energy it receives from the sun.
虽然地球比太阳冷，但这仅仅意味着地球向太空释放的能量比太阳少。我们的星球(一般来说)处于辐射平衡状态，这意味着地球向太空辐射的能量必须等于它从太阳接收到的能量。

2. Which two of these statements is true?

- [ ] Radiowaves, microwaves and X-Rays are all examples of longwave radiation
- [ ] Shortwave radiation (visible light) travels at the speed of light, but radiowaves are slower.
- [x] We can see the sun because it emits a lot of EM radiation as visible light.

解析：With a temperature of around 6000K, much of the sun's electromagnetic radiation is emitted in the visible light bands of the EM spectrum.

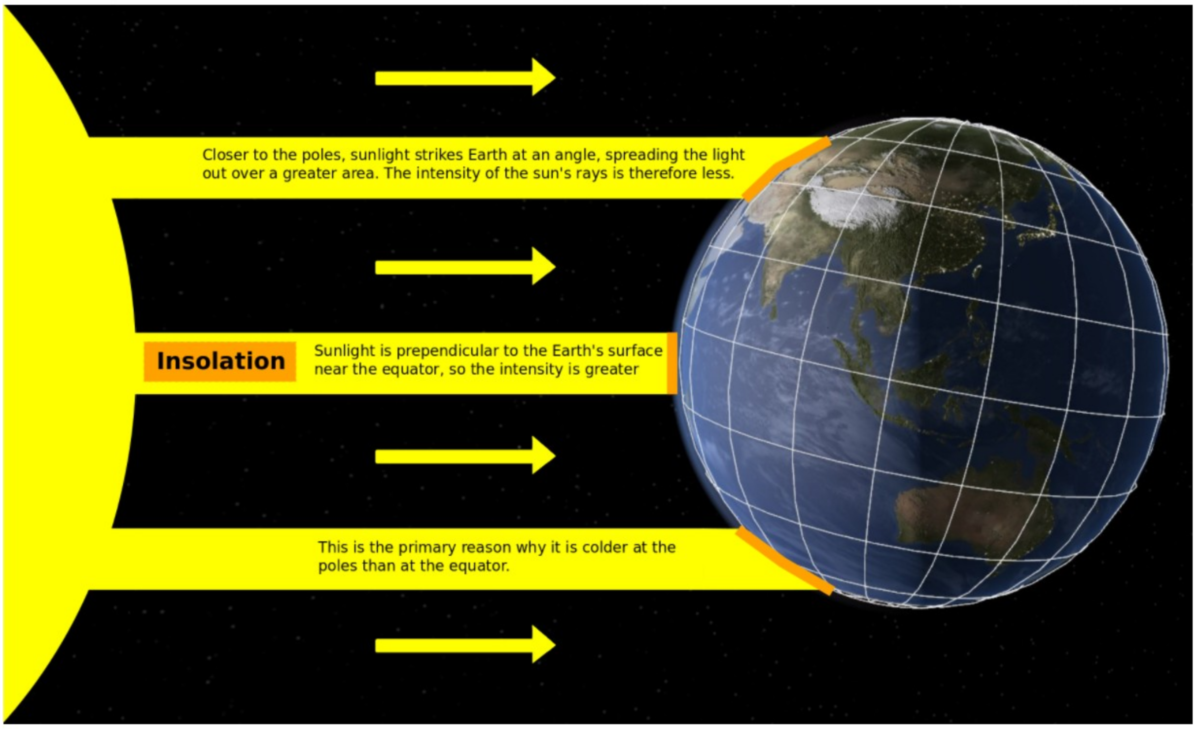
- [x] A blue star will emit more high frequency EM radiation than our yellow sun.

解析：A blue star is hotter than a yellow one and therefore will radiate more EM radiation (according to the Stefan-Boltzman law).

Blue star hotter than yellow hotter than red.

Insolation 日照和地球温度梯度

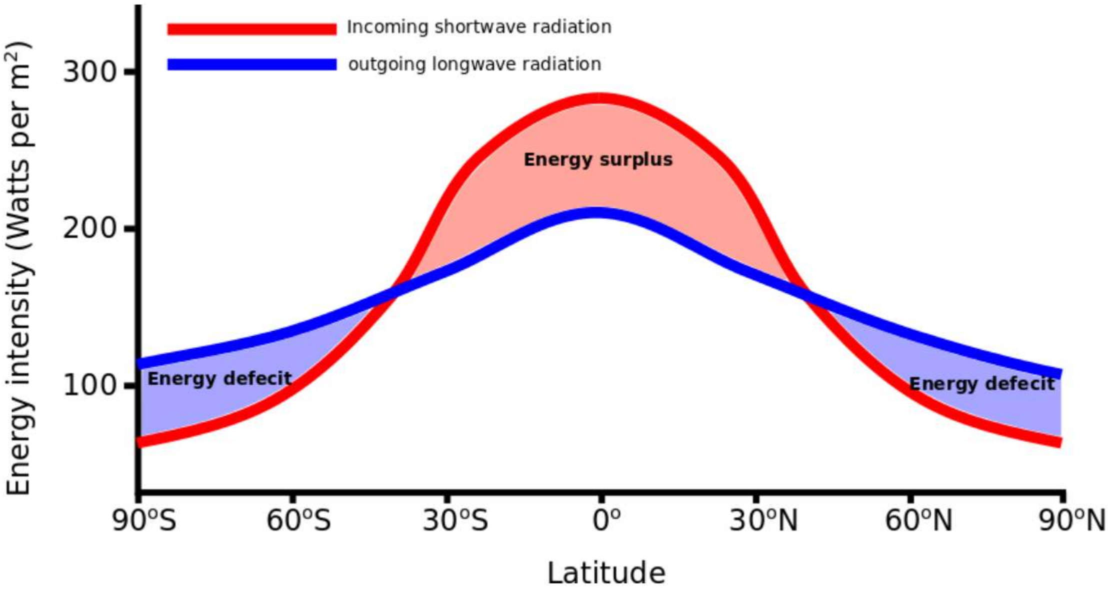
来自太阳的能量(“日照”)并不是以相同的强度照射到地球上所有的地方。这是因为地球(近似)是球形的，所以地球表面垂直于入射太阳光线的地方强度最大。这就是为什么靠近赤道的热带地区日照最充足(见下图)。在靠近两极的地方，阳光以一个角度照射地球，这意味着同样的能量现在分散在一个更大的区域(见底部的两幅图)。



Sunlight has a different intensity at the Earth's surface depending on its angle relative to the Earth's surface. Insolation is most intense near the equator and least intense at the poles.

In addition to the sun's energy heating the Earth, our planet is also constantly losing energy, in the form of radiation of infrared energy back into space. As you learned in the lesson on EM radiation, any object with a temperature above absolute zero (0 K) will radiate electromagnetic radiation. This is also true of the Earth. The shortwave radiation from the sun is absorbed by the atmosphere, land and oceans. All of these 'surfaces' consequently heat up and radiate long-wave energy (*Question: why is the radiation from Earth in the longwave part of the spectrum?*). If we look at the figure below, we can see both the incoming shortwave and outgoing longwave EM radiation as a function of latitude.

除了太阳的能量加热地球，我们的星球也在不断地失去能量，以辐射红外能量的形式返回太空。正如你在电磁辐射课上学到的，任何温度高于绝对零度(0k)的物体都会辐射电磁辐射。地球也是如此。来自太阳的短波辐射被大气、陆地和海洋所吸收。所有这些“表面”因此都会加热并辐射长波能量(问题:为什么来自地球的辐射在光谱的长波部分?)如果我们看下面的图，我们可以看到输入的短波和输出的长波电磁辐射作为纬度的函数。

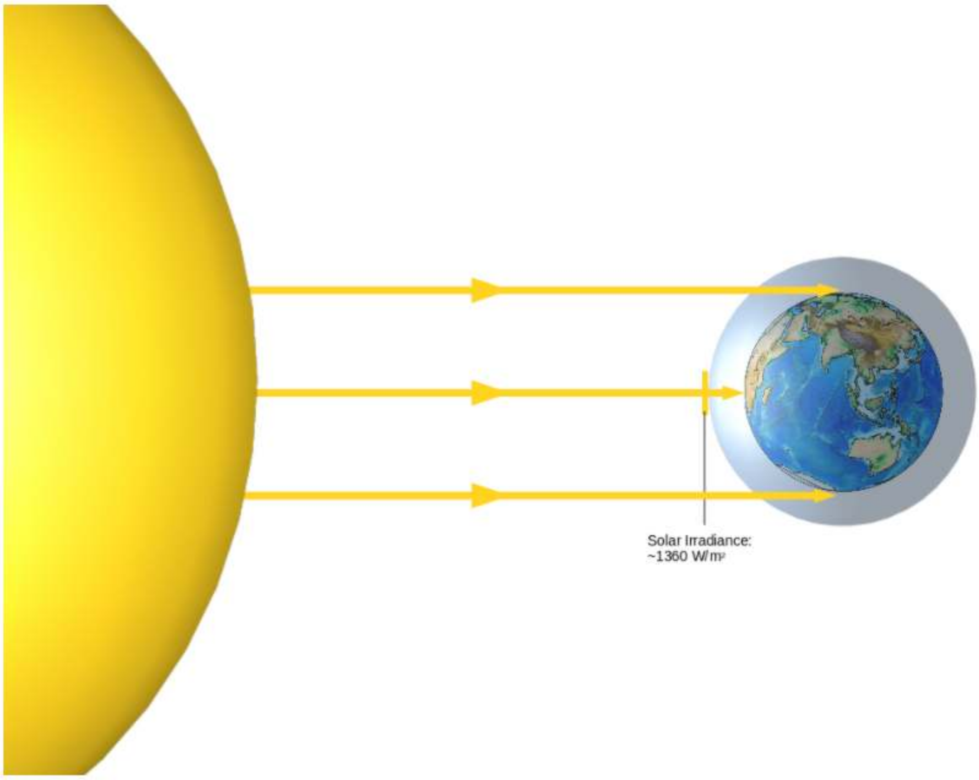


Note that while the energy from insolation (red line) is focused around the lower latitudes, the outgoing longwave radiation (blue line) is a bit more evenly spread out from pole-to-pole. The result of this is a net warming (i.e. an energy and heat surplus) at the equator and a net cooling (i.e. an energy and heat deficit) at the poles. **This is the fundamental reason we have such a dynamic climate system - uneven warming constantly drives excess energy from low latitudes (near the equator) to the high latitudes (the poles).**

请注意，当来自日照的能量(红线)集中在较低的纬度，流出的长波辐射(蓝线)从两极更均匀地扩散。其结果是赤道的净升温(即能量和热量过剩)和两极的净冷却(即能量和热量亏损)。这就是我们有这样一个动态气候系统的根本原因——不均衡的变暖不断地将过剩的能量从低纬度(赤道附近)驱动到高纬度(两极)。

To answer this, we first define the intensity of solar radiation above the Earth's atmosphere, perpendicular to the sun's radiation (see the figure below). For every square metre of area perpendicular to the sun's radiation above the atmosphere, the intensity of solar radiation is about 1360 Watts. This quantity is known as the solar constant (1360 Watts per metre squared, or W/m^2), and can be measured by satellites. Note that the intensity of solar radiation at the Earth's surface is less than this, because some radiation is absorbed by the atmosphere, and most of the Earth's surface is not perpendicular to the sun's radiation.

为了回答这个问题，我们首先定义了地球大气层以上垂直于太阳辐射的太阳辐射强度(见下图)。大气上方垂直于太阳辐射的每平方米面积，太阳辐射的强度约为 1360 瓦。这个量被称为太阳常数(1360 瓦每平方米，或 W/m^2)，可以通过卫星测量。需要注意的是，太阳在地球表面的辐射强度小于这个值，因为一些辐射被大气吸收，而大部分地球表面与太阳的辐射并不垂直。



Also note that only half of the Earth's surface receives sunlight at any one time and also, as we just discussed, this light is not of uniform intensity across this half of the globe. You can imagine that from the sun's perspective, the Earth (being very far away) looks like a disc - indeed it intercepts the same amount of solar radiation as a disc with the same radius. So we can calculate the energy intercepted by the Earth using the area of a circle - πr^2 - and the solar constant.

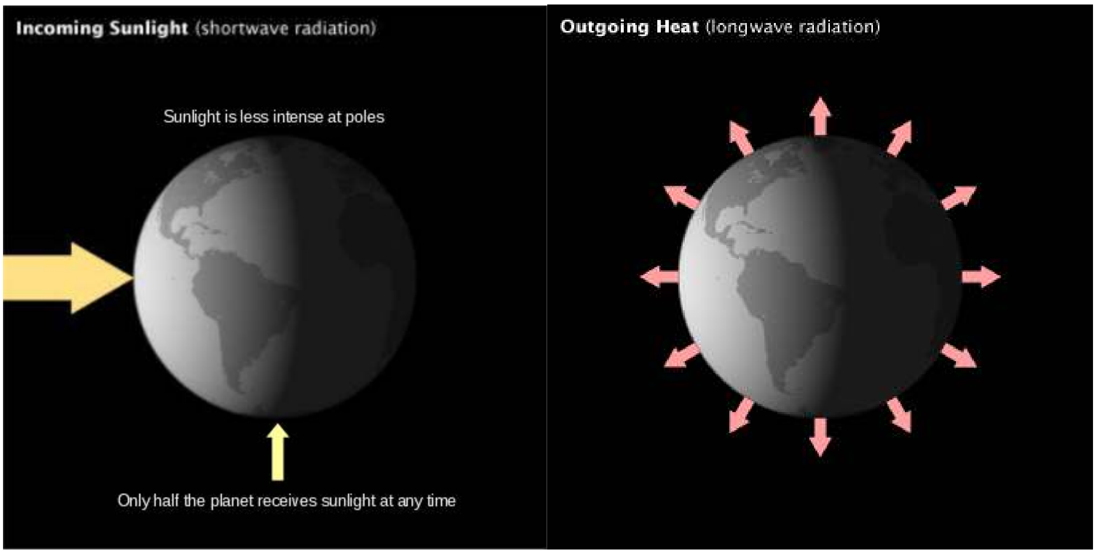
By contrast, the Earth emits longwave radiation in all directions all the time (see figure below, right), over a surface area four times as large as the disc described above (since the surface area of a sphere is $4\pi r^2$).

The amount of longwave radiation leaving the Earth is about the same as the solar shortwave radiation absorbed. If that weren't the case the Earth would warm or cool. We'll pick up on this point in more detail in a later lesson.

还要注意，在任何时候，只有地球表面的一半接收到阳光，而且，正如我们刚才讨论的，在地球的这一半，光线的强度不是均匀的。你可以想象一下，从太阳的角度来看，地球(非常遥远)看起来像一个圆盘——的确，它拦截的太阳辐射量与具有相同半径的圆盘一样。因此，我们可以利用一个圆的面积 πr^2 和太阳常数来计算地球截获的能量。

相比之下，地球一直向各个方向发射长波辐射(见下图右图)，其表面积是上面描述的圆盘的四倍(因为球体的表面积为 $4\pi r^2$)。

离开地球的长波辐射与吸收的太阳短波辐射大致相同。如果不是这样，地球就会变暖或变冷。我们将在以后的课程中更详细地讨论这一点。



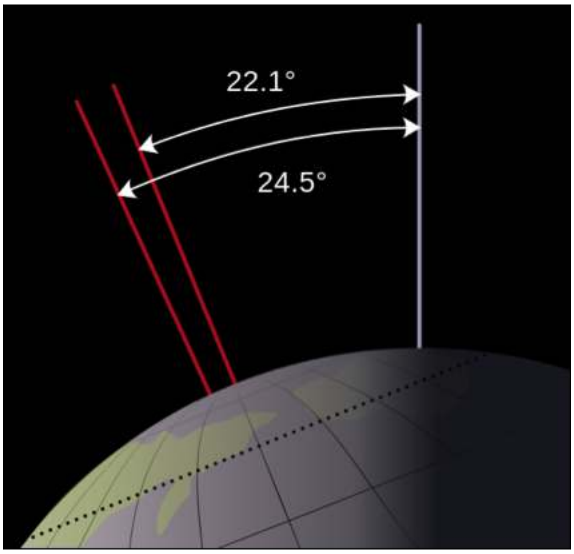
Next we will look at how the Earth's orbit around the sun and its tilt slowly change over time (ranging from tens of thousands to hundreds of thousands of years). We call these Milankovitch cycles, named after Serbian geophysicist and astronomer Milutin Milanković. We'll look at three dominant cycles:

- Obliquity
- Eccentricity
- Precession

接下来我们来看看地球绕太阳公转的轨道和它的倾斜是如何随着时间慢慢变化的(从几万年到几十万年不等)。我们称这些为米兰科维奇周期，以塞尔维亚的地球物理学家和天文学家 Milutin 命名 Milanković。我们将着眼于三个主导周期:

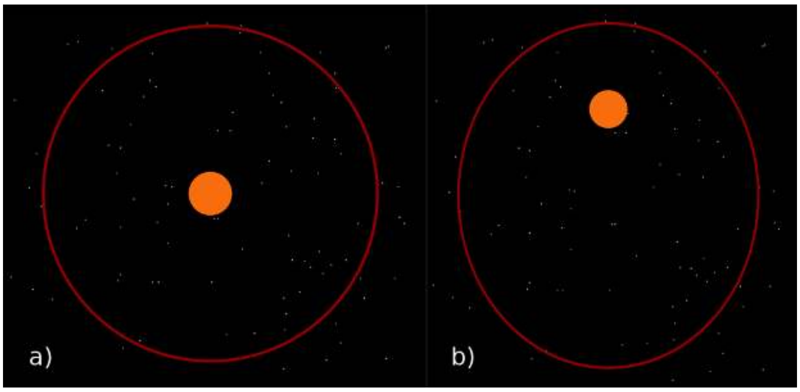
倾角 离心 岁差

The obliquity cycle 倾角周期 simply refers to the change in the axial tilt of the Earth over time. It varies between 22.1 and 24.5 degrees, as shown in the figure above. This cycle has a period of about 41,000 years.



The second cycle we are interested in is the Eccentricity of the Earth's orbit around the sun. Earth's orbit is not actually circular as shown in (a) below, but slightly elliptical (b). The shape of the orbit varies between more circular 圆 and more elliptical 椭圆 on a period of roughly 100,000 years.

我们感兴趣的第二个周期是地球绕太阳轨道的离心率。地球的轨道实际上并不是如图(a)所示的圆形，而是略呈椭圆形(b)。在大约 10 万年的时间里，轨道的形状在更圆和更椭圆之间变化。

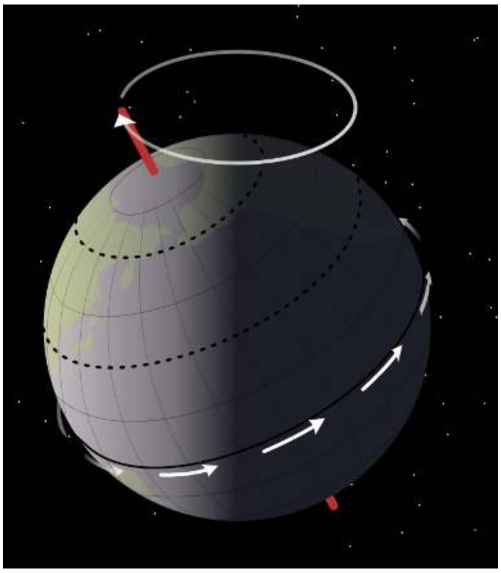


The precession cycle 岁差周期 can be thought of as the slow, circular rotation of the direction in which the Earth's axis points. It has a period of roughly 26,000 years.

Precession causes the timing of the seasons to slowly shift. After half a precession cycle, a point in the Earth's orbit which was summer for one hemisphere will become winter for that hemisphere.

岁差周期可以被认为是地球轴线指向的方向缓慢的圆形旋转。它大约有 26000 年的历史。

岁差导致季节的时间缓慢地改变。在半个岁差周期之后，地球轨道上的某个点对一个半球来说是夏天，对那个半球来说将变成冬天。



维基百科：地球的岁差主要由太阳、月球及其他行星作用在地球赤道隆起部分的引力矩引起。 在天文学和大地测量学中，岁差一般专指地球自转轴缓慢且均匀的变化，周

期约 25,722 年。 其他周期较短或不规律的变化则被称为章动。 岁差的具体表现是地球赤道面和黄道面的变化，这两种变化又分别被称为赤道岁差和黄道岁差。

Sunspot cycles

Finally, we consider changes in the output of the sun itself - which clearly change the amount of energy the Earth receives.

The sun's emission of radiation varies with the roughly 11 year sunspot cycle. This means that the Earth receives either more or less energy from the sun, depending upon the part of the cycle.

Sunspots are dark regions on the photosphere (the outer surface of the sun) about 10,000km in diameter and roughly 1500 degrees C cooler than the surrounding surface. Importantly, these spots are associated with intense magnetic activity and (somewhat paradoxically) result in higher overall solar irradiance.

The sunspot cycle is important to the climate system as it affects the solar constant. Because an increase in sunspots means greater solar activity, this increases the solar constant.

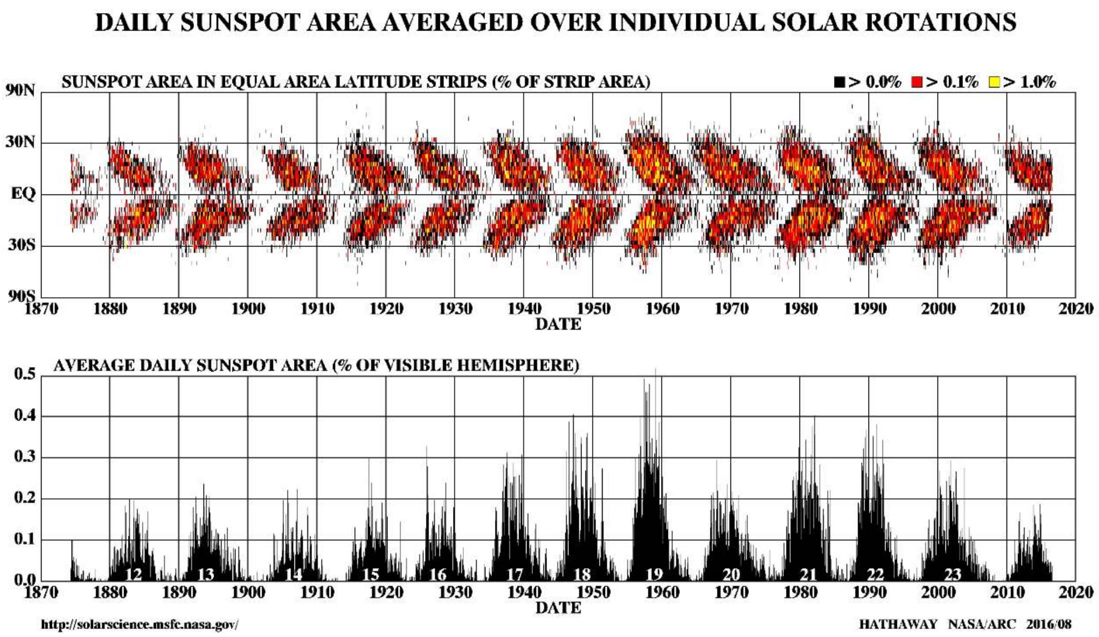
太阳黑子周期

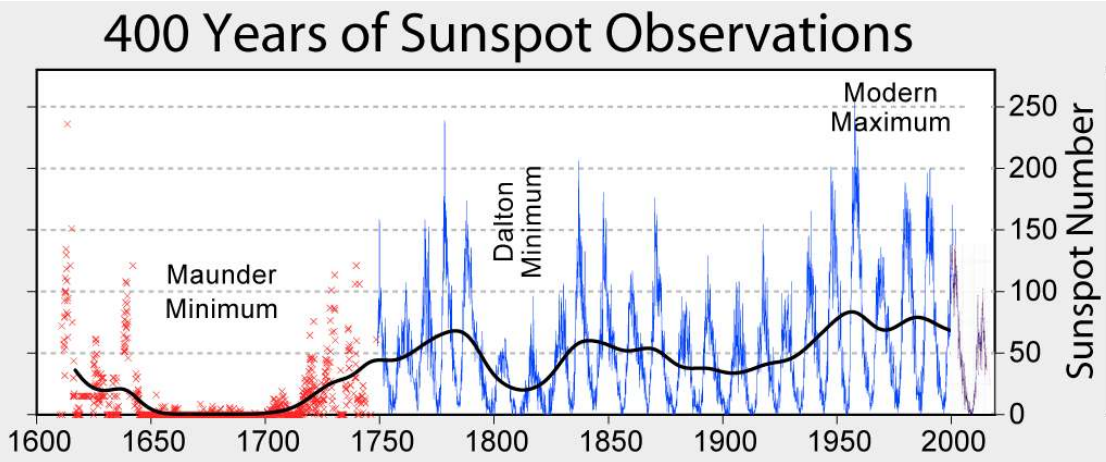
最后，我们考虑太阳本身输出的变化——这显然改变了地球接收到的能量。

太阳辐射的释放随着大约 11 年的太阳黑子周期而变化。这意味着地球从太阳接收到的能量或多或少，取决于循环的部分。

太阳黑子是位于光球层(太阳的外表面)上的黑暗区域，直径约 10000 公里，比周围表面温度低 1500 摄氏度。重要的是，这些黑子与强烈的磁场活动有关，并(有点自相矛盾地)导致更高的太阳总辐照度。

太阳黑子周期对气候系统是重要的，因为它影响太阳常数。因为太阳黑子的增加意味着更大的太阳活动，这增加了太阳常数。





习题

Which of the cycles covered above can change the solar constant? There is more than 1 correct answer.

- [ ] Precession
- [ ] Obliquity
- [x] Eccentricity
- [x] Sunspot Cycle

Solar constant can only be altered in one of two ways:

1 - If the sun's energy output changes or

2 - if the distance between the sun and the Earth changes. Therefore, the correct answers were both eccentricity and the sunspot cycle.

Composition of the Atmosphere

The Earth's atmosphere is primarily made up of just two very common gases (nitrogen and oxygen) and many less common ones (see figure to the left).

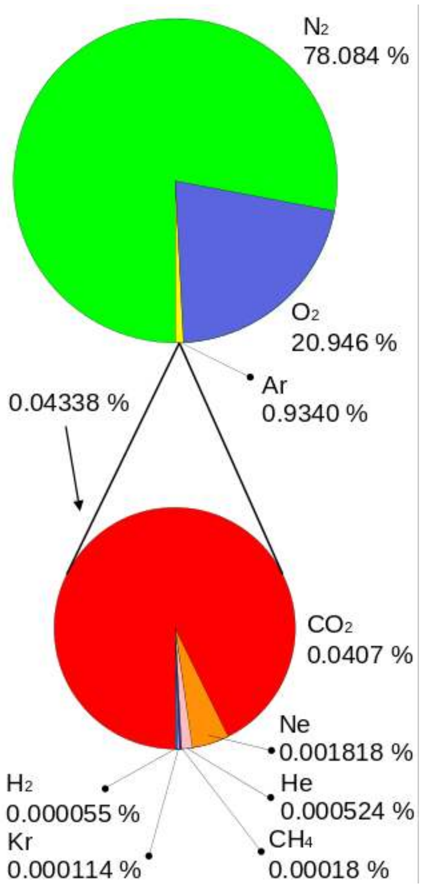
A completely dry parcel of air (meaning without any water vapour in it), is about 78% nitrogen, 21% oxygen and the remainder is primarily argon. There are also many trace gases (those which make up less than 1% of the atmosphere) such as carbon dioxide, methane, helium, hydrogen and more.

The amount of water vapour in the air is highly variable, but it typically makes up about 1% of the atmosphere at any one time.

地球的大气层主要由两种非常常见的气体(氮和氧)和许多不太常见的气体(见左图)组成。

一团完全干燥的空气(即没有任何水蒸气)中，约有 78%是氮气，21%是氧气，剩下的主要是氩。还有许多微量气体(这些气体只占大气的 1%以下)，如二氧化碳、甲烷、氦、氢等等。

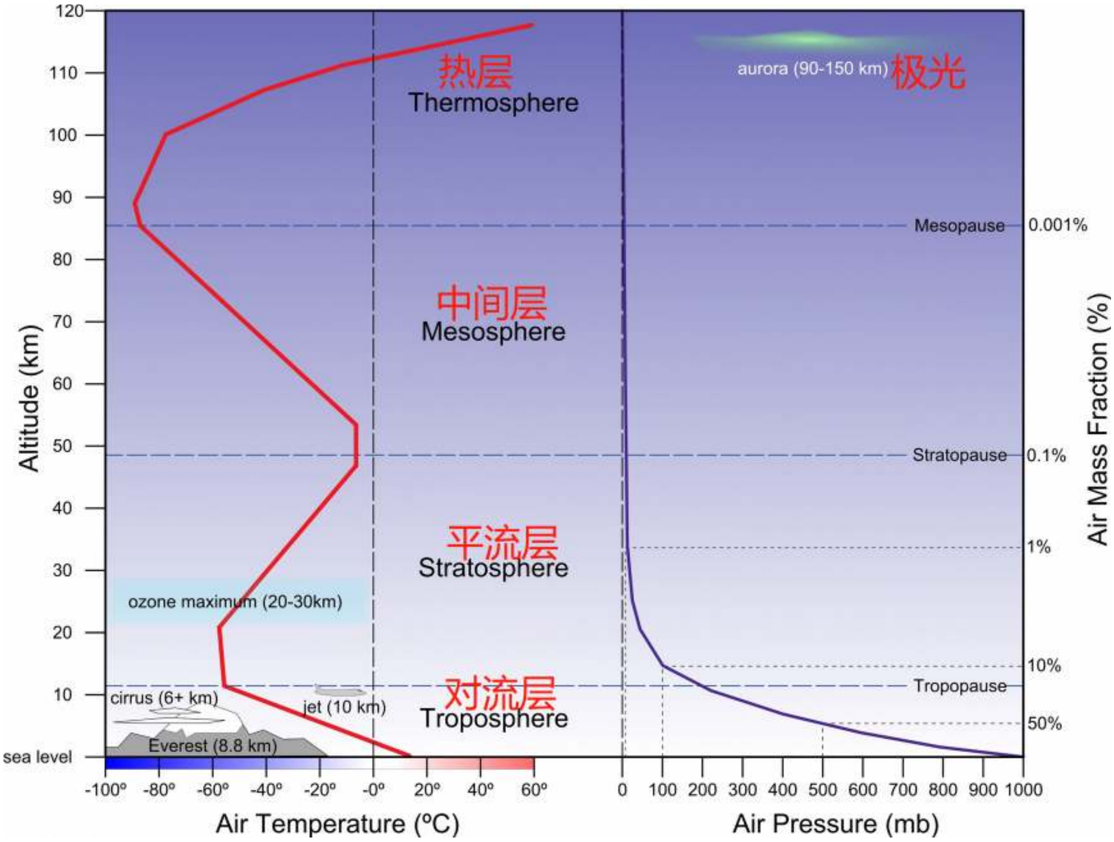
空气中水蒸气的含量变化很大，但它通常在任何时候都占大气的 1%。



Temperature and pressure in the atmosphere

Pressure and temperature are not homogeneous throughout the atmosphere. Most of the mass of the atmosphere lies near the Earth's surface, simply because of the Earth's gravity. In the figure below, you can see that half of the mass of the atmosphere lies below about 5.5km ('50%' on the vertical axis on the right hand side of the figure) and once you reach 16km (in the stratosphere) you are already above 90% of the atmosphere. The solid purple line shows that atmospheric pressure rapidly decreases with height in the lowest section of the atmosphere (the 'troposphere'), then decreases very slowly over the next 100km above the surface. Temperature, however, does not uniformly change with height in the atmosphere. The four named layers of the atmosphere - the troposphere, the stratosphere, the mesosphere and the thermosphere - are based on their temperature characteristics. We'll go through each layer one at a time below.

气压和温度在整个大气中是不均匀的。大气层的大部分质量位于地球表面附近，这仅仅是因为地球的引力。在下面的图中，你可以看到一半的大气质量位于 5.5 千米以下(“50%”在图右手边的纵轴上)，一旦你到达 16 千米(在平流层)，你就已经超过了大气的 90%。紫色实线显示，在大气的最底层(“对流层”)，气压随着高度的增加而迅速降低，然后在离地面 100 公里处，气压又缓慢降低。然而，温度并不会随着大气中的高度一致地变化。四层大气——对流层、平流层、中间层和热层——是根据它们的温度特征命名的。下面我们将逐一介绍每一层。



The troposphere 对流层

The troposphere is where the majority of the mass of the atmosphere lies (about 75%) and also contains 99% of the total atmospheric mass of water vapour and aerosols. It's where the weather that we experience happens and so is generally what people are most interested in. Due to differences in surface heating and atmospheric mixing, the troposphere extends from the surface to about 7km in height at the poles and about 17km near the equator, and also varies in height with seasonal changes. In the troposphere, temperature *decreases with height* as greenhouse gases absorb longwave radiation emitted by the Earth's surface, keeping the lower part of the troposphere warmer than the upper part.

对流层是大气的大部分质量(约 75%)所在的地方，也包含了大气总质量的水蒸气和气溶胶的 99%。它是我们经历的天气发生的地方，所以通常是人们最感兴趣的地方。由于地表加热和大气混合的不同，对流层在两极从地表延伸到高度约 7km，在赤道附近约 17km，高度也随季节变化而变化。在对流层，温度随高度而降低，因为温室气体吸收了地球表面发出的长波辐射，使对流层下部比上部温暖。

The stratosphere 平流层

The stratosphere is the layer above the troposphere and extends from about 17 to 50km in altitude. The stratosphere contains about 20% of the atmosphere's mass and unlike the troposphere, temperature *increases with height* in the stratosphere (see the solid red line in the figure above). This is because of the high concentration of ozone in the stratosphere, which absorbs UV radiation coming from the sun. By the time the sun's shortwave radiation reaches the troposphere, most UV radiation has been absorbed in the stratosphere. Note that UV radiation is linked to skin cancer, genetic damage and immune system suppression in living organisms, so the ozone layer is important! You might be aware of the Antarctic ozone hole that developed in the 1980's - this is why it was a big deal.

平流层是在对流层之上的一层，高度从 17 公里延伸到 50 公里。平流层约占大气质量的 20%，与对流层不同的是，温度随着平流层高度的增加而增加(见上图的红线)。这是因为平流层中有高浓度的臭氧，可以吸收来自太阳的紫外线辐射。当太阳的短波辐射到达对流层时，大部分紫外线辐射已被平流层吸收。请注意，紫外线辐射与皮肤癌、基因损伤和生物体免疫系统抑制有关，所以臭氧层是重要的!你可能知道南极臭氧空洞是在 20 世纪 80 年代形成的——这就是为什么它如此重要。

The mesosphere 中间层

Above the stratosphere is the mesosphere, roughly 50 to 85km above the Earth's surface. At this height, the atmosphere is very thin, and, like in the troposphere, temperature drops as altitude increases due to a weak greenhouse effect (there's not much atmosphere around) culminating in the coldest temperatures found on Earth of around -90$_o$C near the mesopause (the top of the mesosphere). The mesosphere is also where noctilucent clouds form - the highest altitude clouds that can exist.

平流层之上是中间层，距离地球表面大约 50 到 85 千米。在这个高度，大气非常稀薄，和对流层一样，由于有微弱的温室效应(周围没有太多的大气)，温度随着高度的增加而下降，在中顶(中间层顶部)附近达到地球上最冷的温度，约零下 90 摄氏度。中间层也是夜光云形成的地方，这是可能存在的最高海拔的云。

Noctilucent clouds are composed of tiny water crystals at altitudes of around 80 km (in the mesosphere). The clouds are only visible at dusk, when sunlight from below the horizon illuminates them, at high latitudes (above 50°) in both hemispheres.

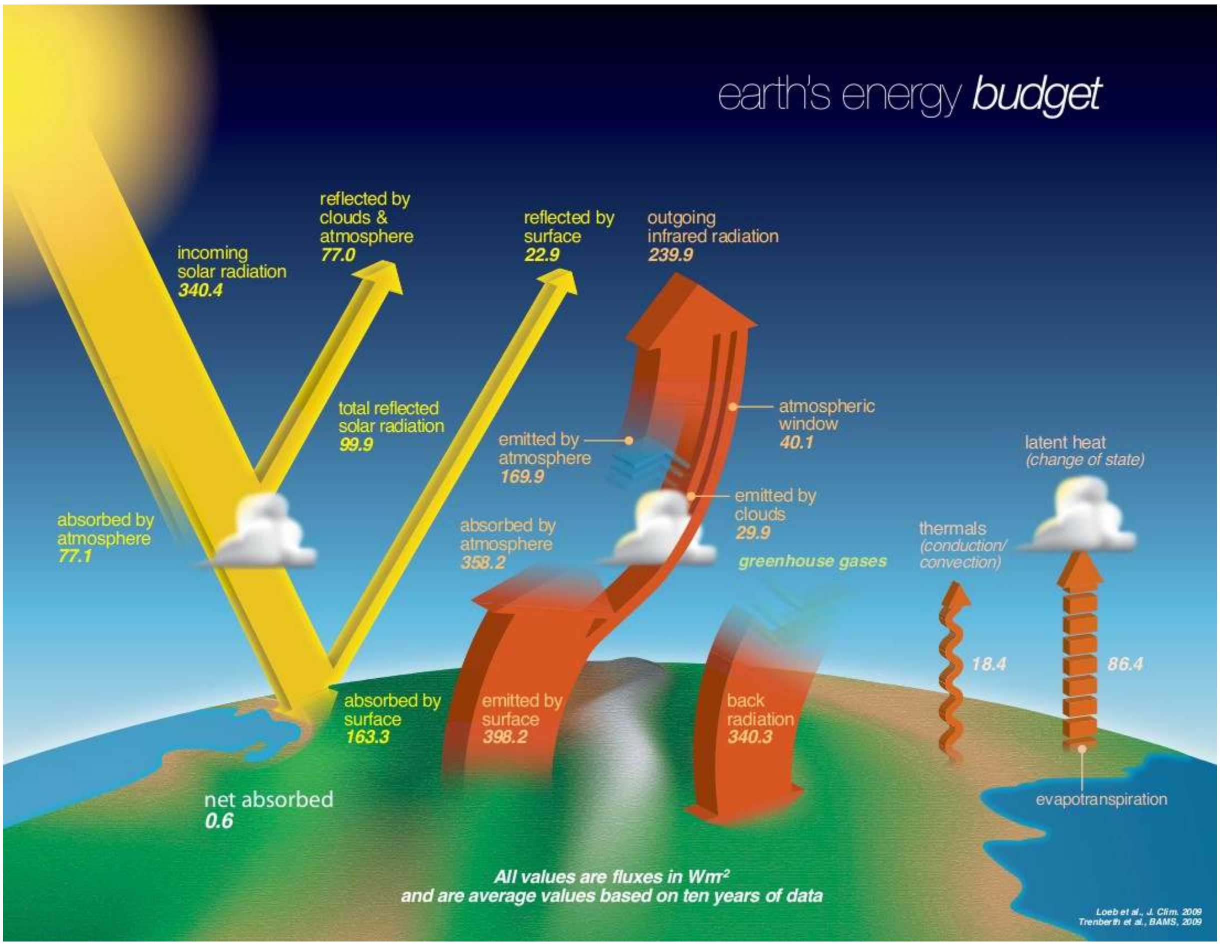
夜光云是由海拔 80 公里左右(在中间层)的微小水晶体组成的。这些云只有在黄昏时才能看到，当阳光从地平线以下照亮它们时，在两个半球的高纬度(50°以上)。

The thermosphere 热层

This is the outermost layer of the atmosphere, extending from about 85km to over 600km (this varies considerably), however, given that the air density is so low, 100km in altitude is generally considered the edge of space. Temperatures steadily *increase* throughout the thermosphere due to the absorption of the shortest wavelength EM radiation from the sun (e.g. x-ray, UV). Even though gases in the thermosphere can reach temperatures of up to 2500^{o}C, you would still feel cold if you were up there, since the atmosphere is so close to being a vacuum.

这是大气的最外层，从大约 85 公里延伸到 600 多公里(变化很大)，然而，考虑到空气密度很低，100 公里的高度通常被认为是空间的边缘。由于吸收了来自太阳的最短波长的电磁辐射(例如 x 射线、紫外线)，整个热层的温度稳步上升。尽管热层中的气体温度可以达到 2500 摄氏度，但如果你在那里，你仍然会感到冷，因为大气非常接近真空。

Solar radiation in the atmosphere



The Earth receives, on average, about 340 W/m^2 of energy from the sun. Around 77 W/m^2 of this is reflected by the clouds and the atmosphere back into space. An approximately equal amount is absorbed by gases such as oxygen, ozone and water vapour (recall from the previous lesson that the stratosphere is warmed by ozone's absorption of UV radiation). Of the radiation that reaches the surface, a further 23 W/m^2 is reflected back into space, leaving around half the initial amount of energy being absorbed by the Earth's surface (that is, 163.3/340.4 in the figure above, or about 48%).

If you're wondering why this figure suggests 340 W/m^2 of incoming solar radiation when the solar constant is around 1360 W/m^2, recall from the Energy to Earth lesson that not all of the Earth receives sunlight at its full intensity. In fact the Earth only receives sunlight at full strength over an equivalent surface area of a disc that is the same radius as the Earth (that is, from the sun, the Earth 'appears' as a disc) - with surface area πr^2. The actual surface area of the Earth is approximately that of a sphere - $4\pi r^2$. So if we wanted to express incoming solar radiation as an average over the the whole Earth, it is approximately the solar constant divided by four: 1360/4 = 340W/m^2. We will *not* assess you on this fact.

地球平均从太阳接收到约 340 瓦/平方米的能量。其中约 77w /m^2 被云层和大气反射回太空。大约等量的臭氧被氧气、臭氧和水蒸气等气体吸收(回想一下上一课，臭氧吸收紫外线辐射使平流层变暖)。在到达地表的辐射中，还有 23w /m^2 被反射回太空，留下约一半的原始能量被地球表面吸收(即上图中的 163.3/340.4，或约 48%)。

如果你想知道为什么当太阳常数约为 1360 W/m^{2}时，这个数字显示的是 340 W/m2 的太阳辐射，请回想一下“能量到地球”这一课，并不是所有的地球都能接收到全部强度的阳光。事实上，地球只有在一个与地球半径相等的圆盘上(也就是说，从太阳来看，地球“看起来”像一个圆盘)接收到充分的阳光，其表面积 πr^2。地球的实际表面积近似于一个球体的表面积 $4\pi r^2$。因此，如果我们想把进入地球的太阳辐射表示为整个地球的平均辐射，它大约是太阳常数除以 4:1360/4 = 340W/m^2。我们不会根据这个事实来评估你。

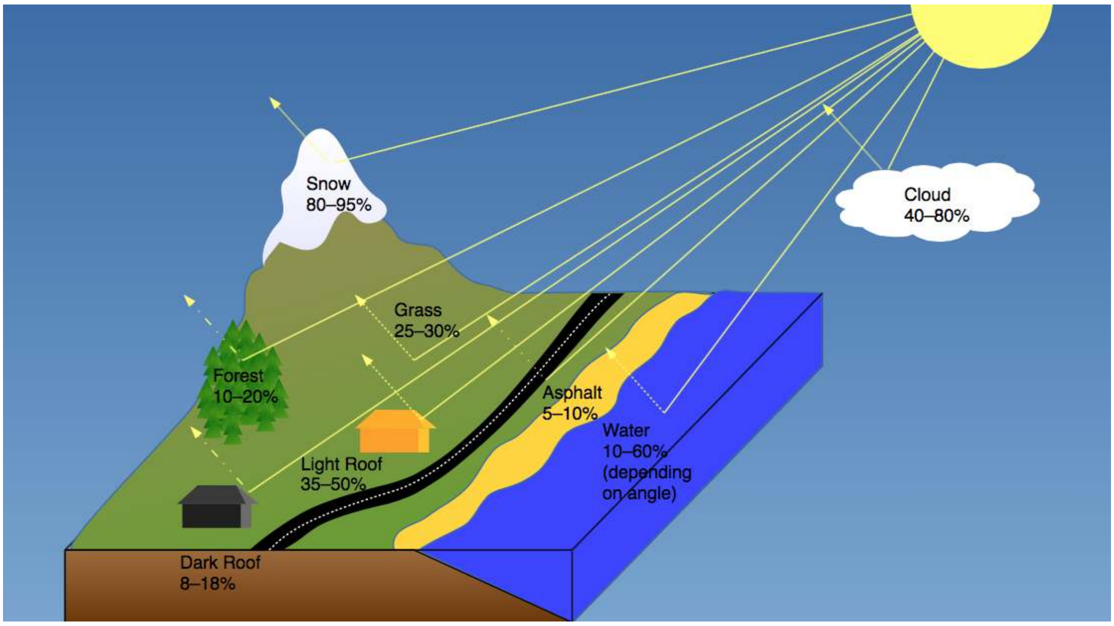
Albedo

Different parts of the Earth system reflect different amounts of solar radiation. Climate scientists use the term 'albedo' to describe the reflectivity of a surface, usually expressed as a number between 0 (absorbs all radiation) and 1 (reflects all radiation) or as a percentage of sunlight reflected.

地球系统的不同部分反映了不同数量的太阳辐射。气候科学家使用术语“反射率”来描述一个表面的反射率，通常表示为一个介于 0(吸收所有辐射)和 1(反射所有辐射)之间的数字，或以阳光反射的百分比表示。

Very white surfaces like snow, ice or cloud have a high albedo and reflect more incoming sunlight, while dark surfaces like roads and buildings have a low albedo and absorb sunlight.

非常白色的表面，如雪、冰或云有很高的反射率，并反射更多的入射阳光，而黑暗的表面，如道路和建筑物有较低的反射率，并吸收阳光。



Aerosols 气溶胶

Aerosols are small (non-gas) particles that are swept around the troposphere and stratosphere by the wind. They range in size from a few nanometres to tens of micrometres and while they're very small in size, they have a significant impact on our climate and our health. Examples of aerosols include organic and black carbon (e.g. smoke from fires or pollens), sulfates (from volcanoes or fossil fuel burning), nitrates, mineral dust, and sea salt. About 90% of aerosols in the atmosphere have natural origins.

气溶胶是一种小型(非气体)颗粒，被风吹过对流层和平流层。它们的大小从几纳米到几十微米不等，虽然它们非常小，但它们对我们的气候和健康有着重大影响。气溶胶的例子包

括有机和黑碳(如火或花粉产生的烟雾)、硫酸盐(来自火山或化石燃料燃烧)、硝酸盐、矿物粉尘和海盐。大气中大约 90%的气溶胶都是自然产生的。

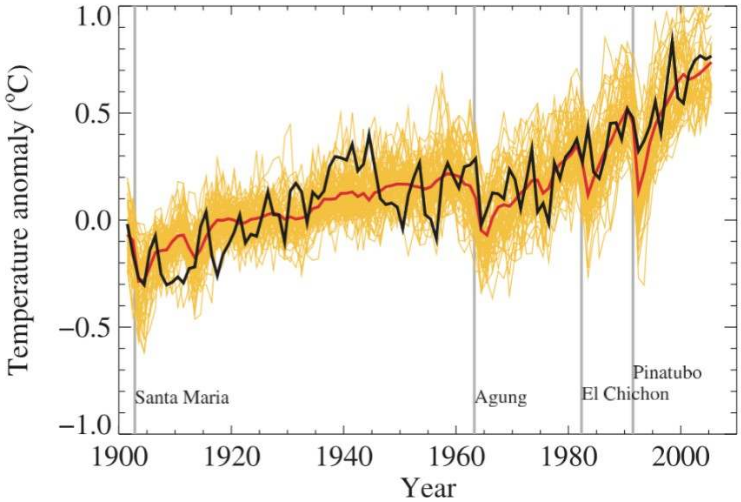
Around 10% of aerosols are of human origin - byproducts of industry, transport, biomass burning and other processes. These aerosols are largely concentrated around urban and industrialised areas and are often most immediately apparent to us as ‘smog’ - the air pollution we see in our cities.

While different aerosols behave differently, with some aerosols absorbing more energy than they reflect, aerosols mostly have a cooling effect on the Earth by reflecting incoming sunlight. We can see their effect on global temperatures in the years after a large volcanic eruption due to the volcanic ash in the atmosphere (see the Figure below showing the effect on global temperature (the black line) of four major eruptions - the vertical grey lines).

大约 10%的气溶胶来自人类——工业、运输、生物质燃烧和其他过程的副产品。这些气溶胶主要集中在城市和工业化地区，我们通常以“雾霾”——我们在城市中看到的空气污染——最为明显。

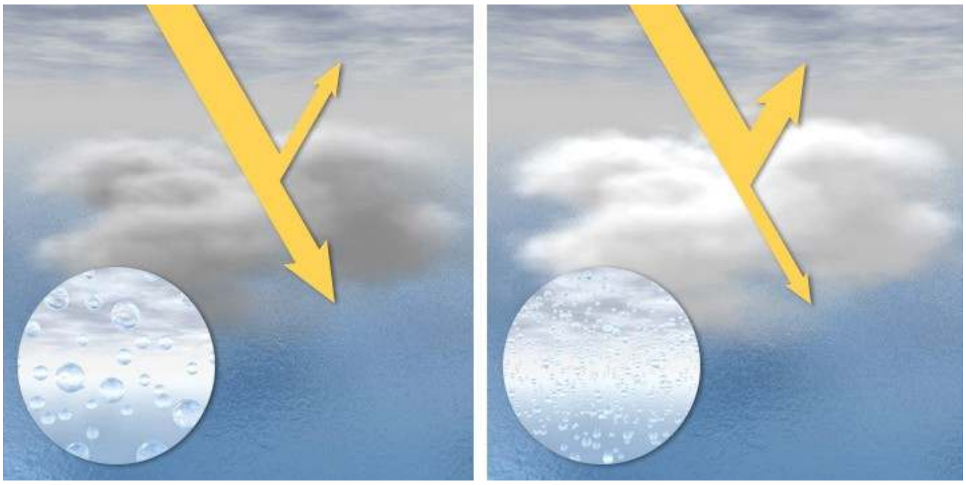
虽然不同的气溶胶行为不同，有些气溶胶吸收的能量比反射的要多，但气溶胶主要通过反射射入的阳光来对地球起到降温作用。我们可以看到它们在一次大型火山爆发后的年份里对全球气温的影响，这是由于大气中的火山灰造成的(见下图显示了四次大型火山爆发对全球气温的影响(黑线)——垂直的灰色线)。

Aerosols also have a secondary effect of increasing the albedo of clouds. They provide more particles in the atmosphere (known as condensation nuclei) for water vapour to condense onto, meaning th



at increases in aerosols lead to more water droplets which in turn leads to optically 'thicker' clouds. Clouds, you will remember, have a high albedo and play a significant role in planetary albedo.

气溶胶还有一个增加云层反射率的次级效应。它们在大气中提供了更多的粒子(被称为凝结核)，让水蒸气凝结在上面，这意味着气溶胶的增加会导致更多的水滴，反过来又会导致光学上“更厚”的云。你会记得，云有很高的反照率，在行星反照率中起着重要的作用。



The image on the left shows a 'normal' state in the atmosphere where some sunlight is reflected by the cloud while some passes through. The image on the right shows a cloud that has formed in air which is rich in aerosols. Note all the extra water droplets in this cloud. This makes the cloud more reflective, so less sunlight passes through it to reach the ground.

左边的图像显示了大气中的一种“正常”状态，一些阳光被云层反射，而一些阳光穿过云层。右边的图像显示了在空气中形成的云，富含气溶胶。注意这朵云中所有多余的水滴。这使得云层反射性更强，因此通过云层到达地面的阳光更少。

Scattering 散射

So far we've seen that incoming shortwave radiation can be absorbed by gases in the atmosphere (e.g. UV absorption by ozone) and reflected by clouds and aerosols. These interactions prevent a proportion of solar radiation from reaching the surface. A third possible pathway for this radiation that might prevent it reaching the surface is scattering, where solar radiation interacts with either gas molecules or aerosol particles. There are two kinds of scattering that are most relevant here:

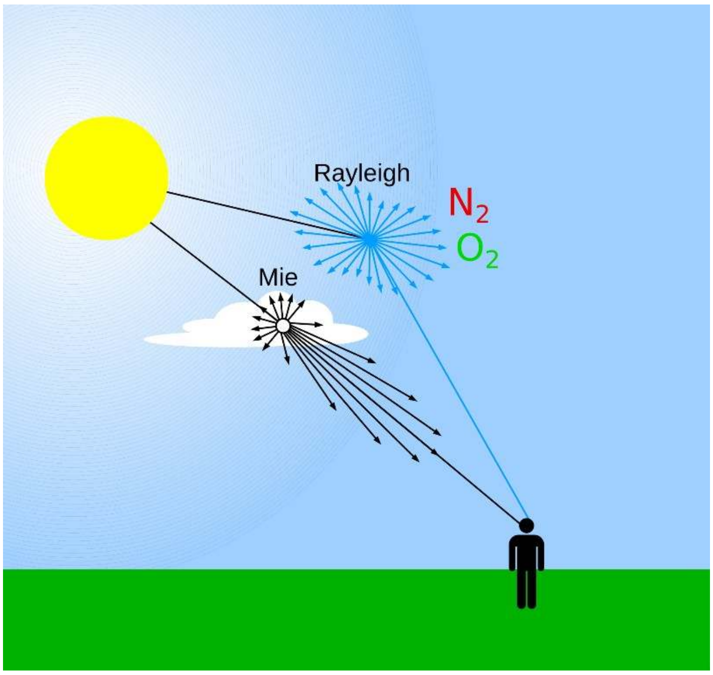
1. **Rayleigh scattering:** When sunlight enters the atmosphere, it scatters due to interaction with nitrogen and oxygen molecules which are much smaller than the wavelength of the radiation. This scattering effect is known as Rayleigh scattering and is particularly strong at the shorter wavelengths of the EM spectrum. This is why red wavelengths can pass through the atmosphere more easily, while blue wavelengths are scattered far more effectively - this gives the sky its blue colour.
2. **Mie scattering:** Mie scattering occurs when sunlight interacts with particles of a similar size or larger than the wavelength of incoming EM radiation, and unlike Rayleigh scattering, it is no more or less efficient at different wavelengths. Mie scattering is also stronger in one direction (see figure below), unlike Rayleigh scattering which propagates radiation relatively evenly in all directions. Mie scattering is responsible for the white haze we often see around clouds, fog, dust or smoke in the air.

到目前为止，我们已经看到，进入大气层的短波辐射可以被大气中的气体吸收(例如臭氧吸收紫外线)，并被云层和气溶胶反射。这些相互作用阻止一部分太阳辐射到达地表。第三种

可能阻止辐射到达地表的途径是散射，即太阳辐射与气体分子或气溶胶粒子相互作用。这里有两种最相关的散射:

瑞利散射:阳光进入大气层时，由于与氮和氧分子的相互作用而散射，这些分子比辐射的波长小得多。这种散射效应被称为瑞利散射，在短波长的电磁光谱中尤其强烈。这就是为什么红色的波长更容易穿过大气层，而蓝色的波长散射得更有效——这就赋予了天空蓝色。

米氏散射:米氏散射发生于阳光与入射电磁辐射波长相近或更大的粒子相互作用时。与瑞利散射不同的是，米氏散射在不同波长上的效率并不高或低。米氏散射在一个方向上也更强(见下图)，不像瑞利散射在所有方向上传播辐射相对均匀。米氏散射是造成白色雾霾的原因，我们经常看到周围的云，雾，灰尘或烟雾在空气中。

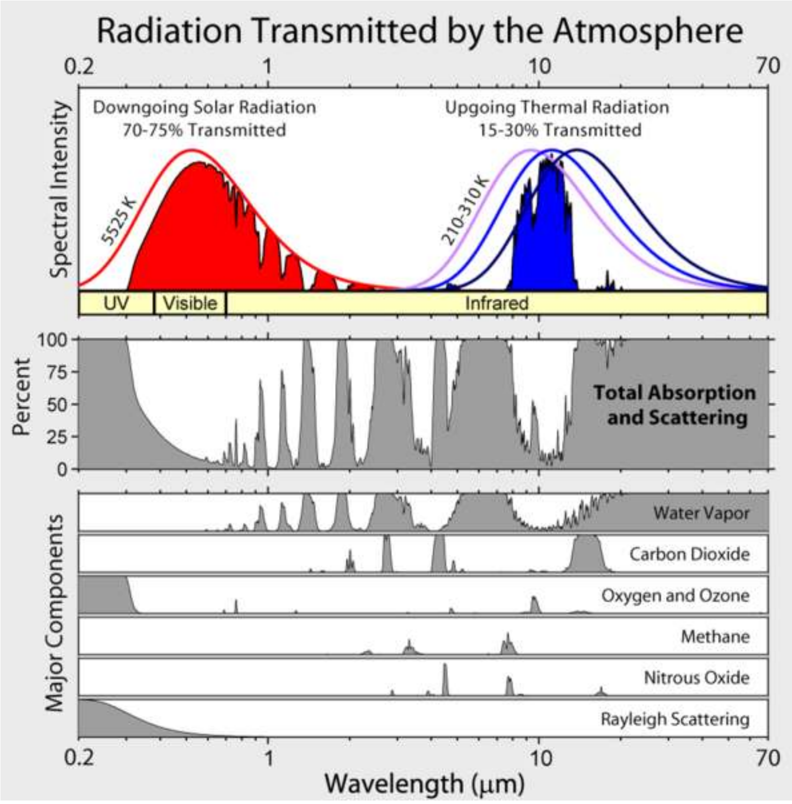


While the atmosphere on a clear day is relatively transparent for the wavelengths of sunlight, the same is not true for the longwave radiation emitted by the Earth. As you can see from the figure above, around 90% (358.2/398.2) is absorbed by the atmosphere, and only 10% directly escapes to space ('atmospheric window' in the figure above). As a result of absorbing this energy, the atmosphere heats up and re-radiates energy in all directions, including back down towards the surface ('back radiation' in the figure above). This keeps the Earth's surface much warmer than it would otherwise be, and is called the *greenhouse effect*. The gases in the atmosphere that absorb and re-radiate longwave radiation are called greenhouse gases.

The figure below shows a representation of the emission spectrum of both the Sun (red line) and the Earth (blue line) as well as which gases in the atmosphere are able to absorb energy in different wavelengths (grey shaded sections below). The shaded red and blue sections give an indication of how much radiation in each wavelength is transmitted through the atmosphere. As noted in the previous lesson, ozone is effective at absorbing high energy UV radiation. You can also see that water vapour, carbon dioxide, methane, nitrous oxide, and ozone are greenhouse gases (since they absorb radiation). Water vapour is responsible for most of the greenhouse effect on Earth (36-72%), next is carbon dioxide (9-26%) then methane (4-9%) then ozone (3-7%).

虽然晴天的大气对阳光的波长来说是相对透明的，但地球发出的长波辐射就不是这样了。从上图可以看出，大约 90%(358.2/398.2)被大气吸收，只有 10%直接逃到太空(上图中的“大气窗口”)。由于吸收了这些能量，大气加热并向各个方向重新辐射能量，包括向地面反射(上图中的“反射辐射”)。这使得地球表面比不这样做时要温暖得多，这就是所谓的温室效应。大气中吸收并再辐射长波辐射的气体称为温室气体。

下图展示了太阳(红线)和地球(蓝线)的发射光谱，以及大气中哪些气体能够吸收不同波长的能量(下面灰色阴影部分)。红色和蓝色的阴影部分显示了每个波长通过大气传输的辐射量。如上一课所述，臭氧能有效吸收高能紫外线辐射。你还可以看到，水蒸气、二氧化碳、甲烷、一氧化二氮和臭氧都是温室气体(因为它们吸收辐射)。水蒸汽是造成地球温室效应的主要原因(36-72%)，其次是二氧化碳(9-26%)，甲烷(4-9%)，然后是臭氧(3-7%)。



For Earth's climate to be stable, the incoming energy from the sun must be equal to the outgoing energy emitted by the Earth. If emitted energy was not equal to incoming energy, the Earth would either gain or lose energy - get hotter or colder. As noted in the video at the beginning of the lesson, the greenhouse effect has been around for billions of years and is part of Earth's radiative equilibrium. In the top figure you can see that most of the energy that the Earth loses to space comes from longwave radiation from the greenhouse gases in the atmosphere ('outgoing infrared radiation'), rather than the surface.

为了使地球的气候稳定，来自太阳的能量必须等于地球发出的能量。如果发射的能量与入射的能量不相等，地球就会得到或失去能量——变热或变冷。正如这节课开始时的视频所指出的，温室效应已经存在了数十亿年，是地球辐射平衡的一部分。在上图中，你可以看到，地球损失给太空的大部分能量来自大气中的温室气体的长波辐射(“流出的红外辐射”)，而不是地表。

So we've seen that the composition of the atmosphere makes the surface of our planet is much warmer than it might otherwise be. Water vapour, which is responsible for the largest part of the greenhouse effect, is variable in space and time, and people have very little direct impact on this. We saw in the last lesson that carbon dioxide (CO_2) makes up only 0.04% of the atmosphere, yet it is responsible for as much as a quarter of the greenhouse effect. To see how effective CO_2 is as a greenhouse gas, we can look at other planets in our solar system.

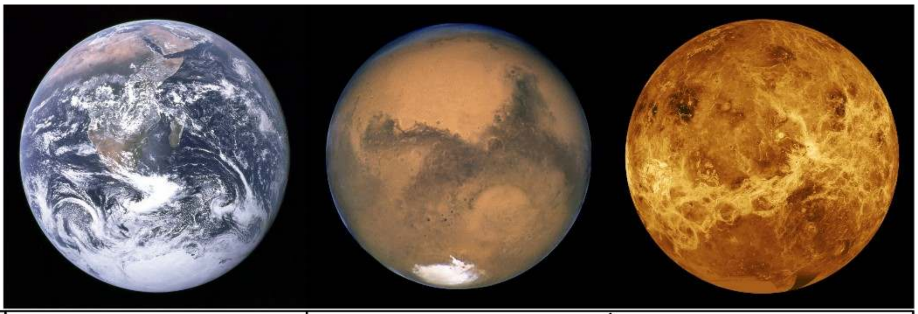
Venus is an excellent example of a greenhouse effect. Its atmosphere has 93 times as much mass as that of Earth and 96% of it is CO_2. Its atmosphere is so thick that, even though the planet as a whole receives almost twice as much sunlight as the Earth (due to being closer to the sun), its surface receives less sunlight than Earth's, as the atmosphere reflects about 75% of it back into space. So even though the surface receives less energy from the sun, the incredibly thick atmosphere of primarily CO_2 creates searing temperatures on the planet's surface (see table below), hotter even than on Mercury (which is much closer to the sun).

On Mars, even though the atmosphere is also largely (96%) made up of CO_2, it is much thinner than on Earth (about 0.5% of the mass of our atmosphere), so while what is there is an excellent greenhouse gas, there is not enough of it to keep the planet very warm.

我们已经看到，大气的组成使得地球表面比正常情况下要暖和得多。造成温室效应最大部分的水蒸气，在时间和空间上都是可变的，而人们对它的直接影响很小。我们在上节课中看到，二氧化碳(CO2)只占大气的 0.04%，但它却造成了高达四分之一的温室效应。为了了解二氧化碳作为温室气体的有效性，我们可以看看太阳系的其他行星。

金星是温室效应的一个很好的例子。它的大气质量是地球的 93 倍，96%是二氧化碳。它的大气层是如此之厚，即使整个行星接收到的阳光是地球的两倍(由于离太阳更近)，它的表面接收到的阳光却比地球少，因为大气层将大约 75%的阳光反射回了太空。因此，尽管火星表面接收到的来自太阳的能量更少，但以二氧化碳为主的极其厚重的大气层却在火星表面造成了灼热的温度(见下表)，甚至比水星(离太阳近得多)还要高。

在火星上,即使气氛也很大程度上(96%)组成的二氧化碳,它比地球上更薄(大约 0.5%的我们的大气层的质量),因此,虽然是有一个优秀的温室气体,没有足够的让这个星球上很暖和。



Planet	Earth	Mars	Venus
Mass of atmos. (% of Earth)	100	0.49	9300
CO_2 concentration %	0.04	96.0	96.5
Avg. surface temp °C	15	-55	462